(A) Reactions which can form aldehydes and Ketones (Both)

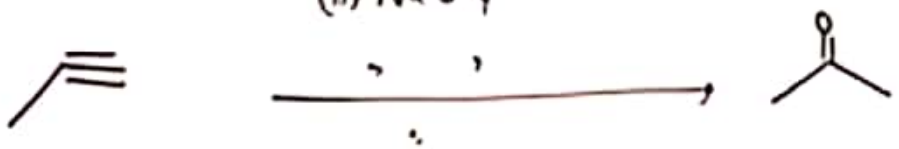
(A1) By hydration of alkyne:- (in ORM-II)

$$H-C\equiv C-H + H_2O \xrightarrow{H_2SO_4 + HgSO_4} CH_3-CH=O$$



A2 By oxy-mercuration demercuration of alkyne (in ORM-II)

$$H-C\equiv C-H \xrightarrow[(ii)\ NaBH_4]{(i)\ H_2O + Hg(OAc)_2} CH_3-\overset{O}{\overset{\|}{C}}-H$$



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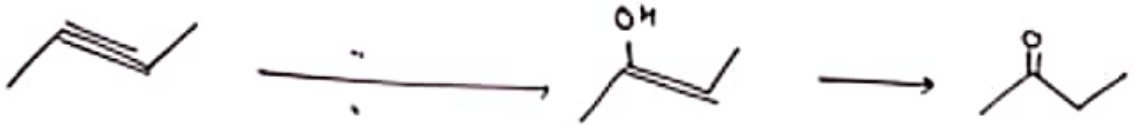
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CARBONYL COMPOUNDS

(A) Reactions which can form aldehydes and Ketones (Both)

(A-3) By hydroboration-oxidation of alkyne (in ORM-II)





A-4 By Oxidation of 1°, 2° alcohols:- (Red. oxide)

$$R-CH_2-OH \xrightarrow[\text{(WOA)}]{\text{PCC / PDC / Colin's Reagent / Oppanaurs ox.}} R-\overset{O}{\overset{\|}{C}}-H$$

$$R-\underset{OH}{\overset{1^\circ}{CH}}-R \xrightarrow{\text{(WOA) or Jones or SOA}} R-\overset{O}{\overset{\|}{C}}-R$$

A-5 By dehydrogenation of 1°, 2° alcohols with Cu, Δ :-

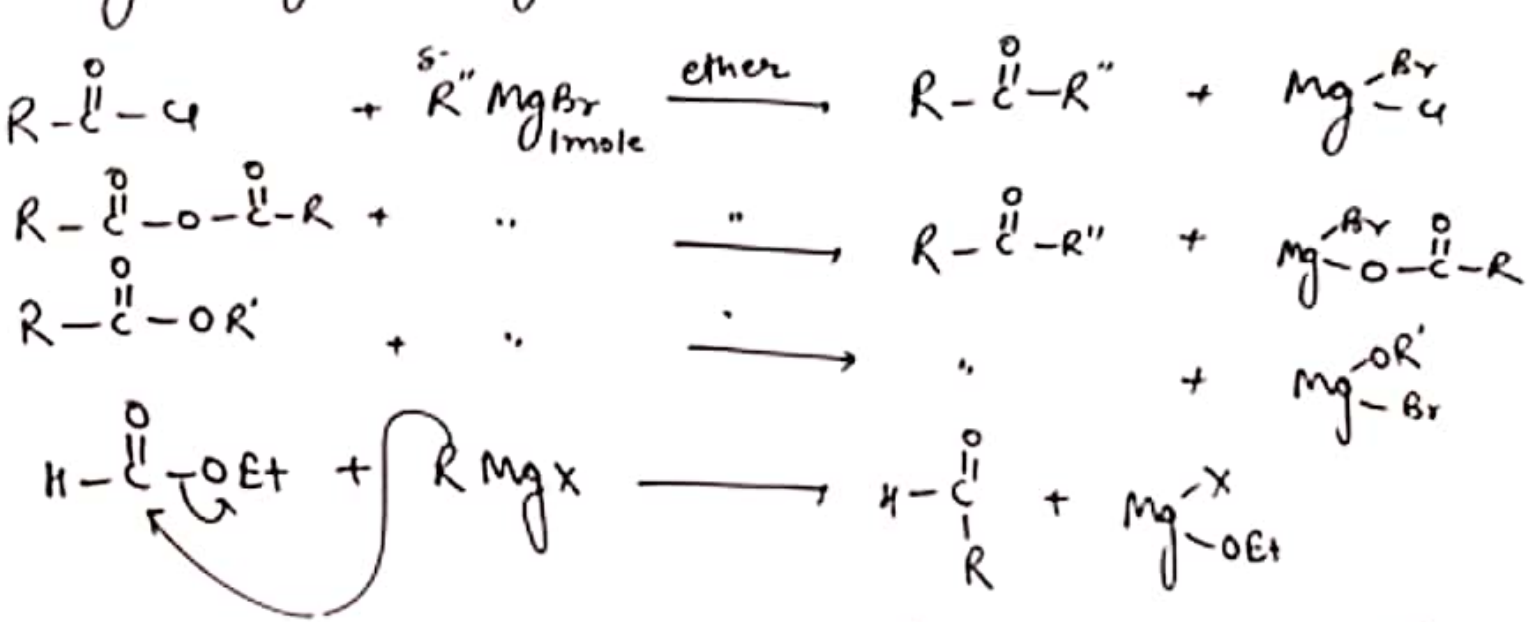
$$1^\circ\ R-CH_2-OH \xrightarrow{Cu,\ \Delta} R-CHO \quad / \quad 2^\circ\ R-\underset{OH}{CH}-R \xrightarrow{Cu\ \Delta} R-\overset{O}{\overset{\|}{C}}-R$$

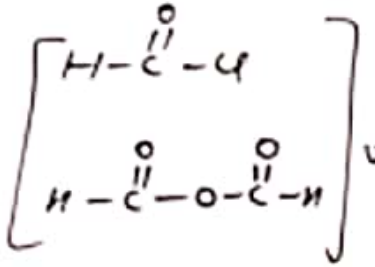
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CARBONYL COMPOUNDS

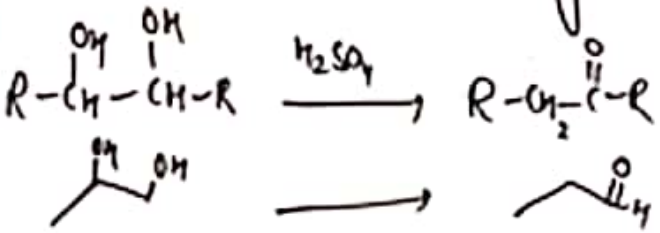
(A) Reactions which can form aldehydes and Ketones (Both)

A.6 By Grignard Reagent:- (in ORM-I S_N2)





A.7 By Pinacol-Pinacolone rearrangement of vic. diol (in ORM-IX)

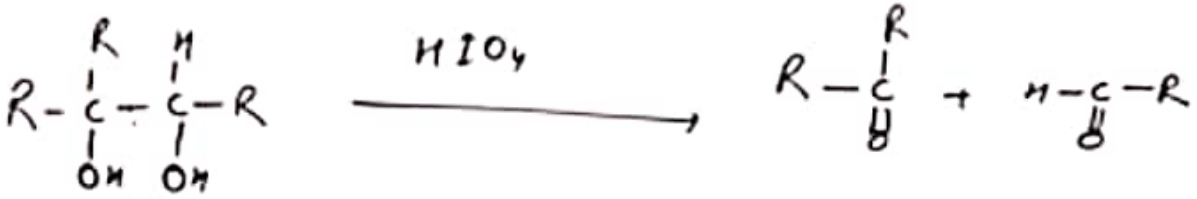


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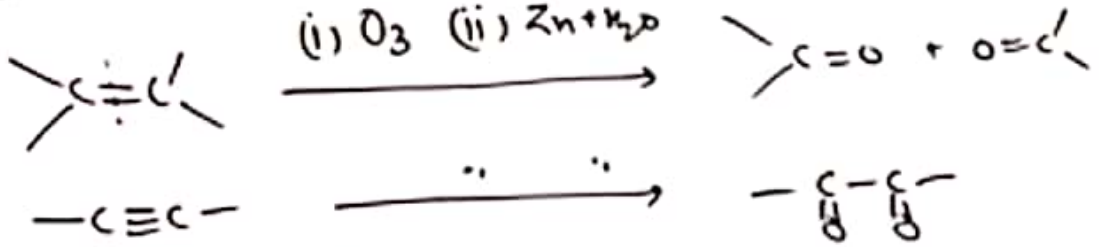
CARBONYL COMPOUNDS

(A) Reactions which can form aldehydes and Ketones (Both)

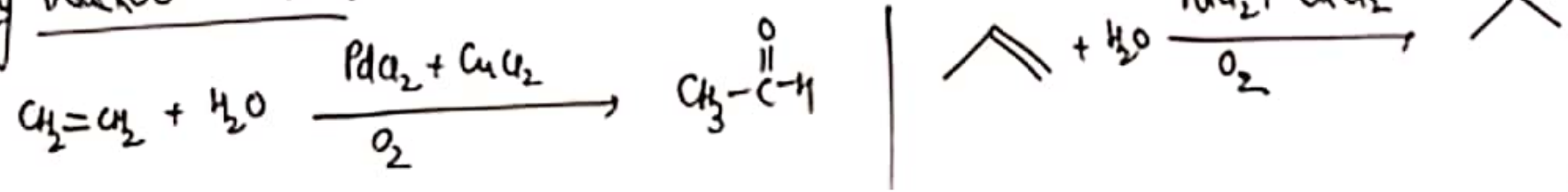
A.8 By Oxidation of vic diol with HIO_4 or $(CH_3COO)_4Pb$.



A.9 By Reductive ozonolysis of alkenes & alkynes:



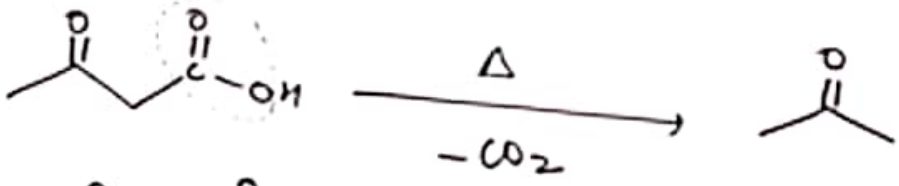
*A.10 By Wacker's Process:-

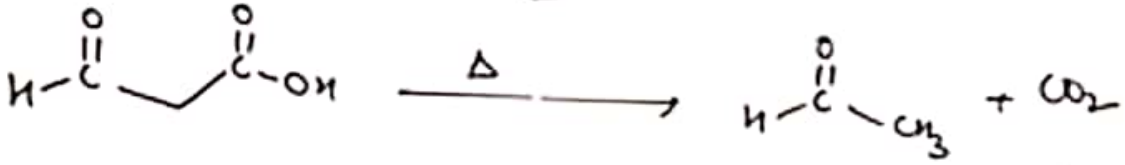


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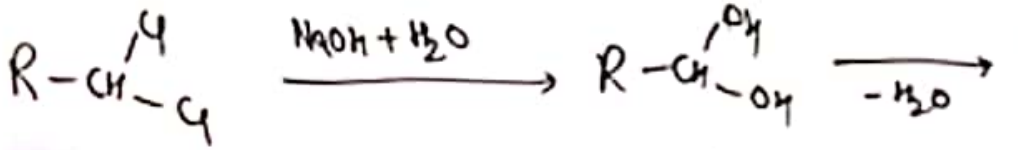
(A) Reactions which can form aldehydes and Ketones (Both)

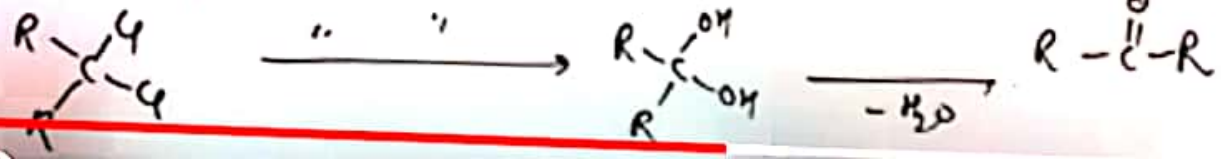
A11 By heating of β-Keto acids:- (decarboxylation)





A12 By Hydrolysis of geminal dihalides (ORM-III)



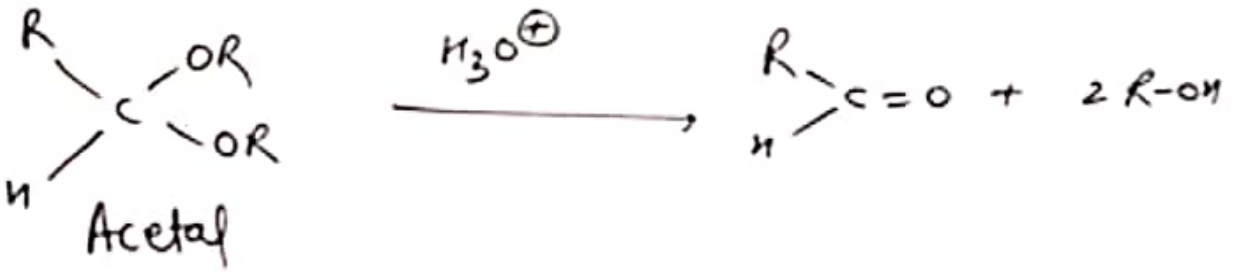


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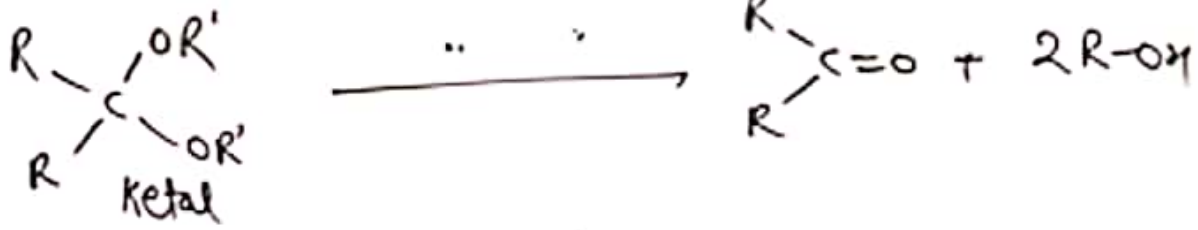
CARBONYL COMPOUNDS

(A) Reactions which can form aldehydes and Ketones (Both)

(A13) By hydrolysis of Ketal & acetal :-

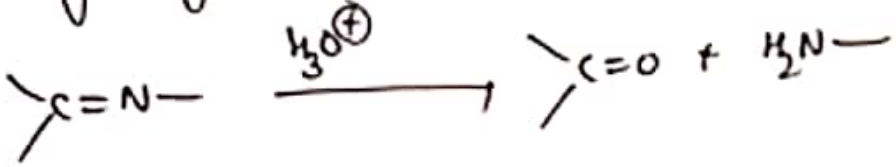


Acetal



Ketal

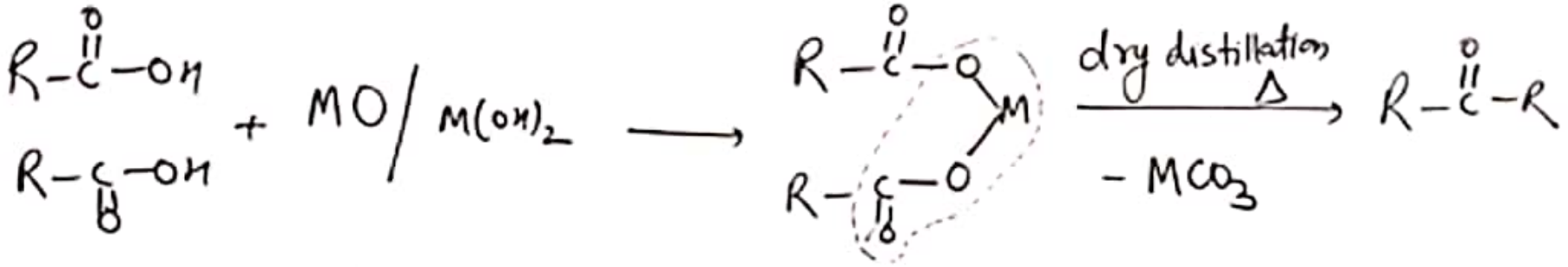
A.14 By hydrolysis of Imine:-



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(A) Reactions which can form aldehydes and Ketones (Both)

A-15 By dry distillation of Ca/Mn/Ba salts of carboxylic acid.



(M = Ca, Ba, Mn)

eg1 $(H-COO)_2Ca \xrightarrow{\Delta} H-CHO + CaCO_3$

eg2 $(CH_3COO)_2Ca \longrightarrow CH_3-CO-CH_3 + CaCO_3$

eg3 $Ph-COOH \xrightarrow{MnO\ \Delta} Ph-CO-Ph$

eg4 $(CH_3COO)_2Ca + (H-COO)_2Ca \xrightarrow{\Delta} \xrightarrow{+CO_2}$ $H-CHO$, CH_3-CHO

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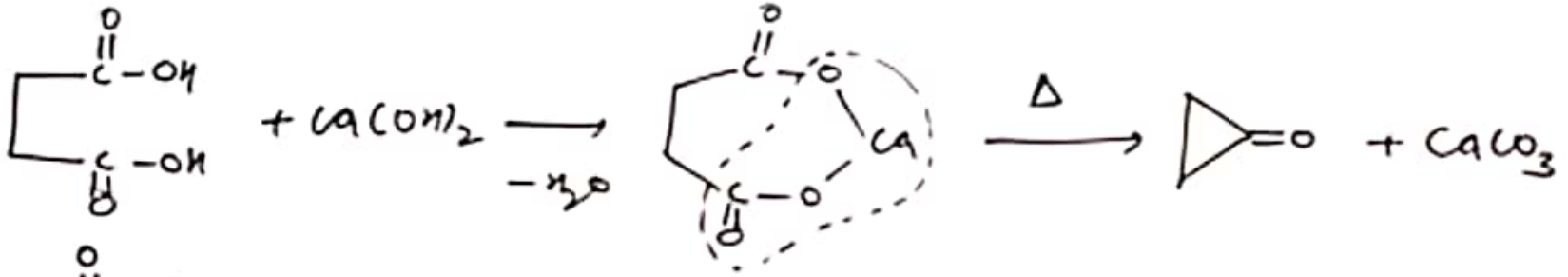
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CARBONYL COMPOUNDS

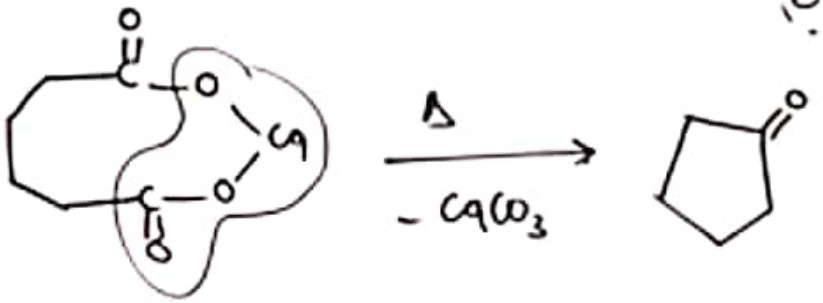
(A) Reactions which can form aldehydes and Ketones (Both)

A·15 By dry distillation of Ca/Mn/Ba salts of carboxylic acid.

eg 5



eg 6



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CARBONYL COMPOUNDS

(B) Reaction for formation of aldehydes only :-

(B.1) By Stephen's Reduction of cyanide (Reduction)

$$R-C\equiv N \xrightarrow{Sn+HCl\,/\,SnCl_2+HCl} R-\overset{O}{\overset{\|}{C}}-H + NH_4Cl$$

(B.2) By Rosenmunds Reduction of acid halide

$$R-\overset{O}{\overset{\|}{C}}-Cl + H_2 \xrightarrow[\text{Rosenmund's reduction}]{Pd + BaSO_4 + (S/X_y/S)} R-\overset{O}{\overset{\|}{C}}-H$$

$$Pd + CaCO_3 + S/X_y$$

not Rosenmund's reduction

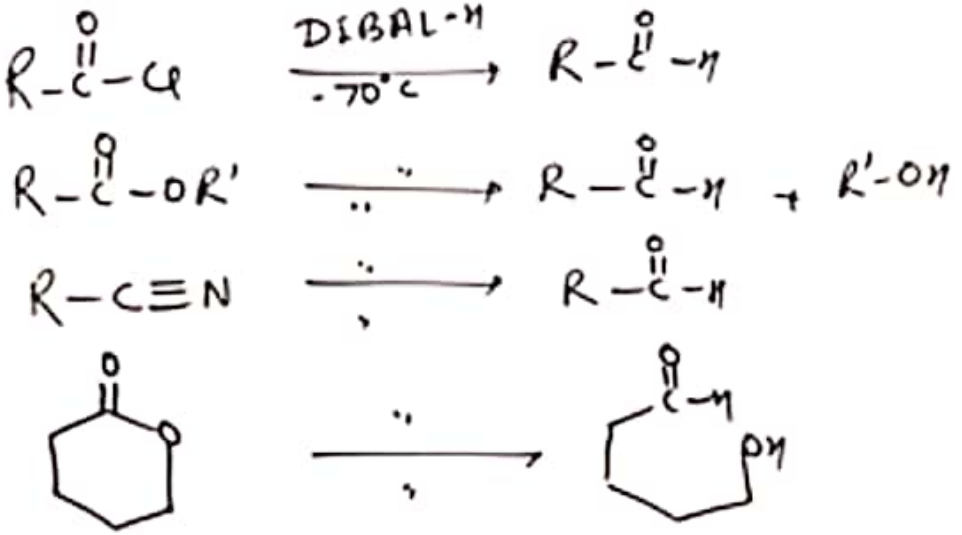
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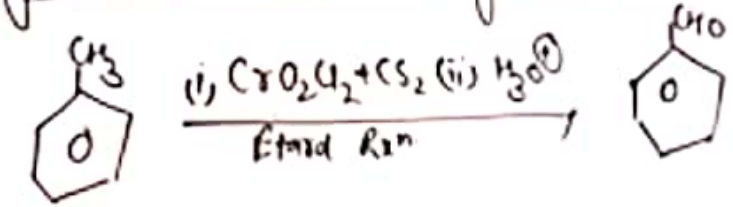
CARBONYL COMPOUNDS

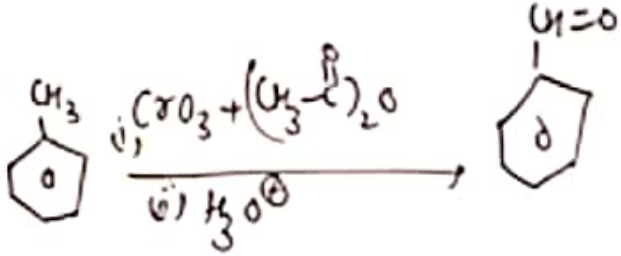
(B) Reaction for formation of aldehydes only :-

B.3 By Partial Reduction of , RCOCl, RCOOR and R-CN with DIBAL-H at −70°C



(B.4.) By Etard Oxidation of toluene :-

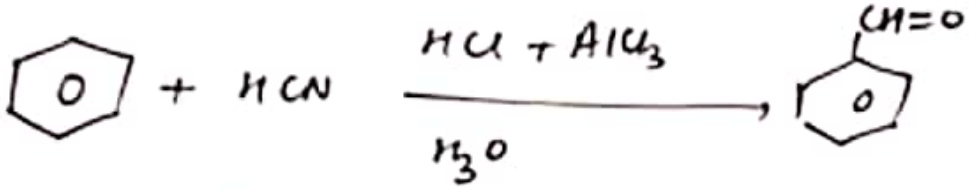




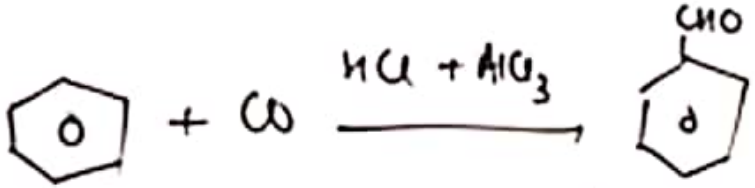
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(B) Reaction for formation of aldehydes only :-

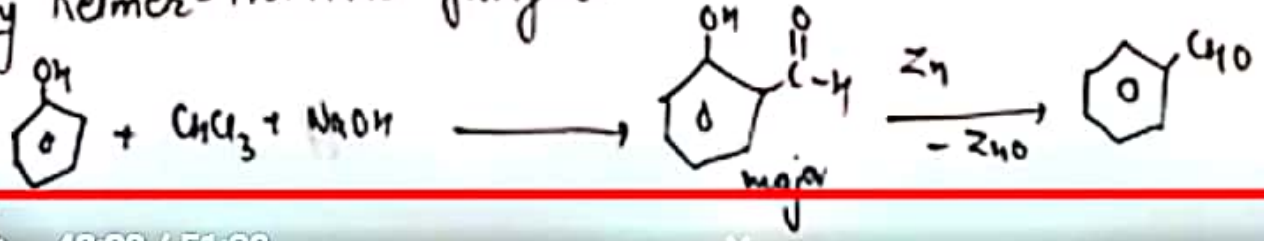
B5 By Gattermann's reaction :- (formylation)



B6 By Gattermann's Koch-formylation



B7 By Reimer-Tiemann's formylation :-



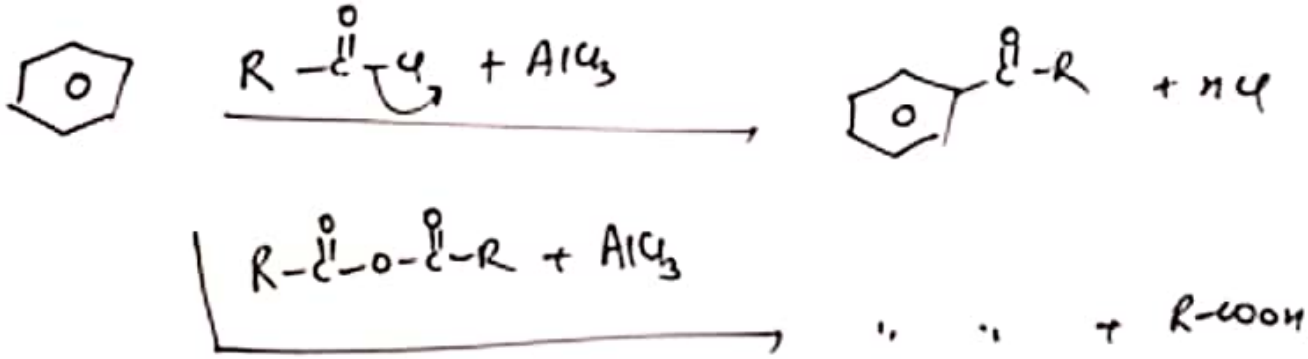
48:29 / 51:28

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CARBONYL COMPOUNDS

[C] Reactions for formation of Ketones only :-

C.1 By Friedel craft acylation



C.2 By Reactions of Gilmann's Reagent (R_2CuLi) with R–COCl :-

$$R-\overset{O}{\overset{\|}{C}}-Cl + R'_2CuLi \xrightarrow{SN^2\ Th} R-\overset{O}{\overset{\|}{C}}-R' + R'Cl + LiCl$$

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CARBONYL COMPOUNDS

Chemical Reactions:-

[A] Reduction & Oxidation ✓

[B] Nucleophilic Addition Reaction.

[C] Reactions of α. acidic hydrogens / enolate ions.

[D] Other Reactions.

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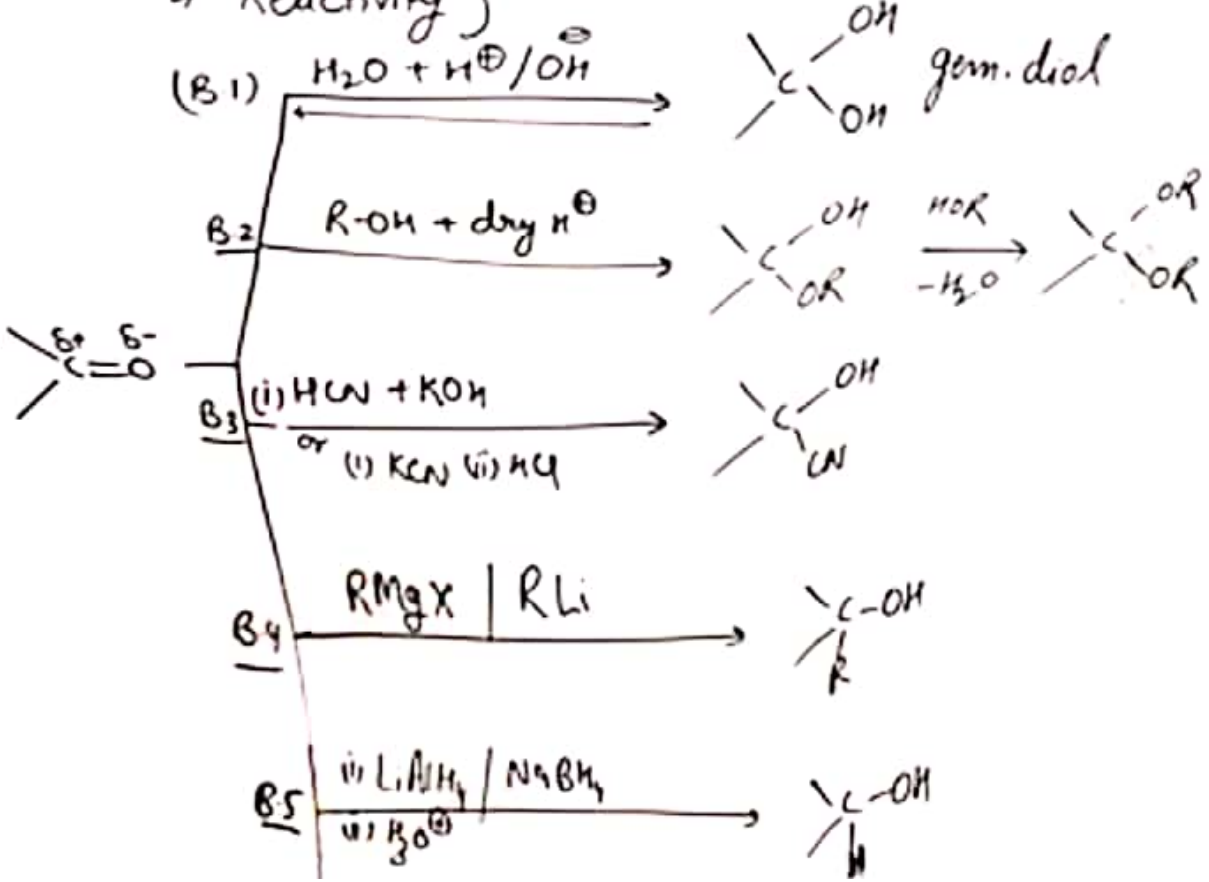
CARBONYL COMPOUNDS

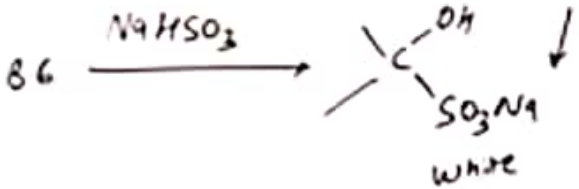
Chemical Reactions:-

[B] Nucleophilic Addition Rxn:-

- Mechanism
- Reactivity

} ORM-I





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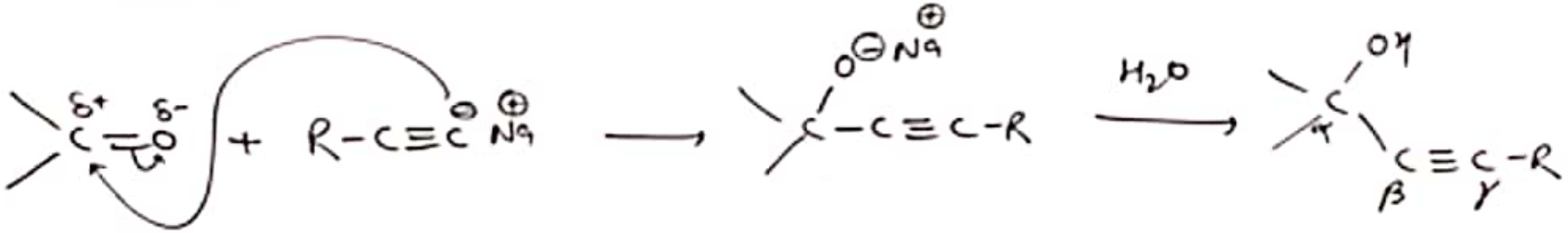
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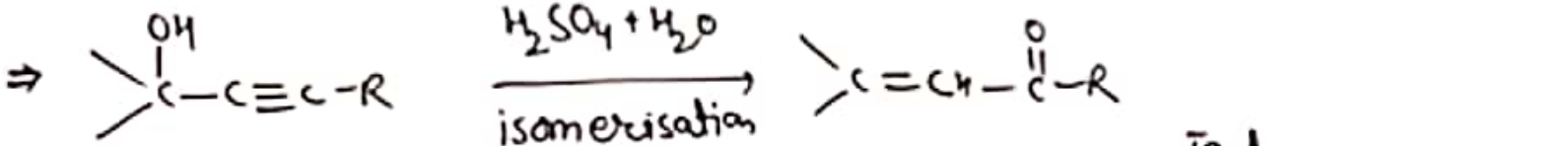
CARBONYL COMPOUNDS

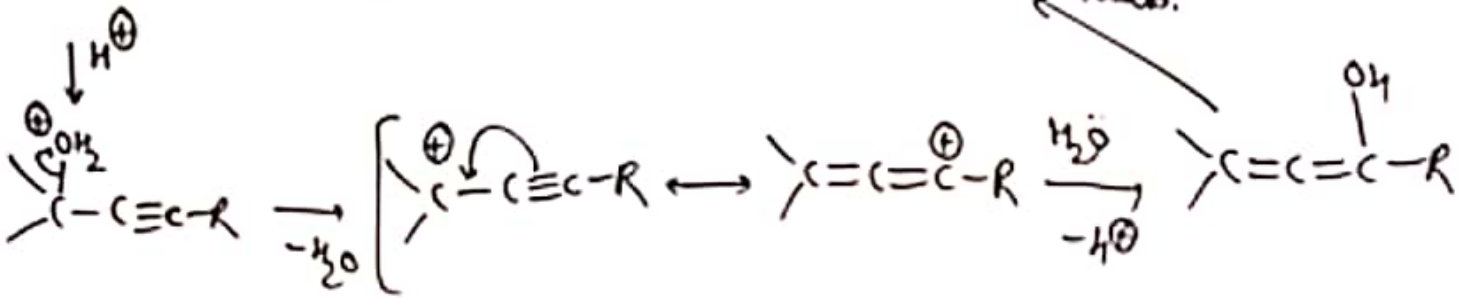
Chemical Reactions:-

[B] Nucleophilic Addition Rxn:-

B.7 Reaction with alkylides







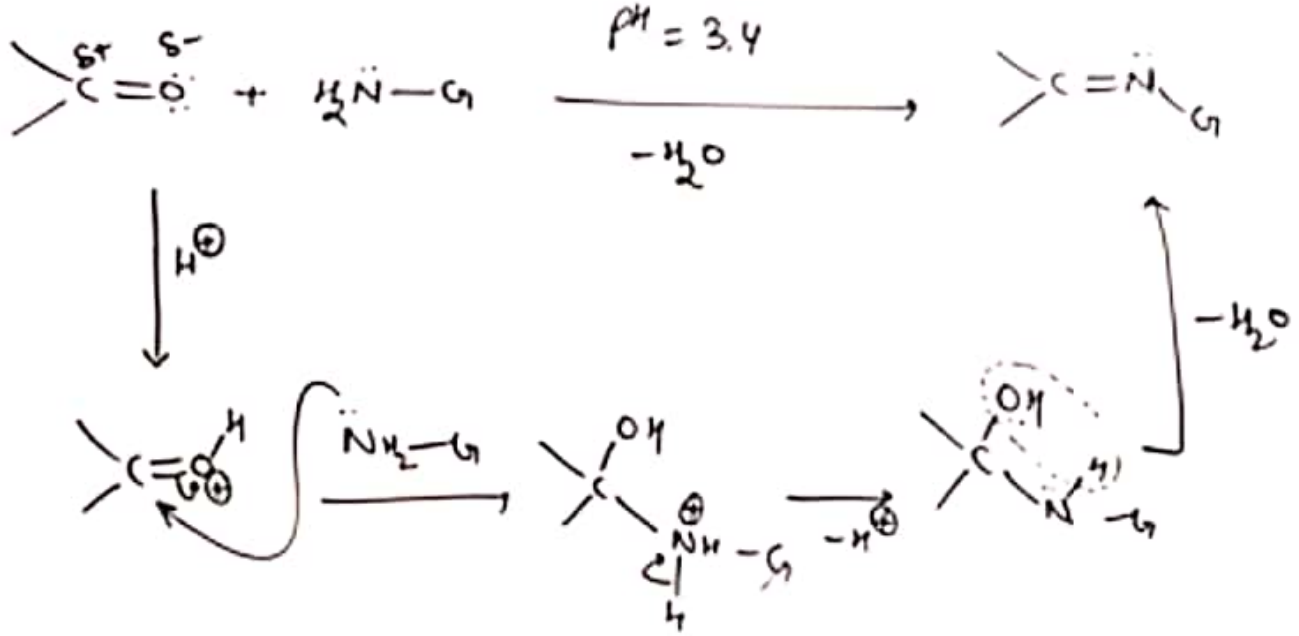
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CARBONYL COMPOUNDS

Chemical Reactions:-

[B] Nucleophilic Addition Rxn :-

B8 Reactions with N containing Nucleophile:- (Addition - Elimination)



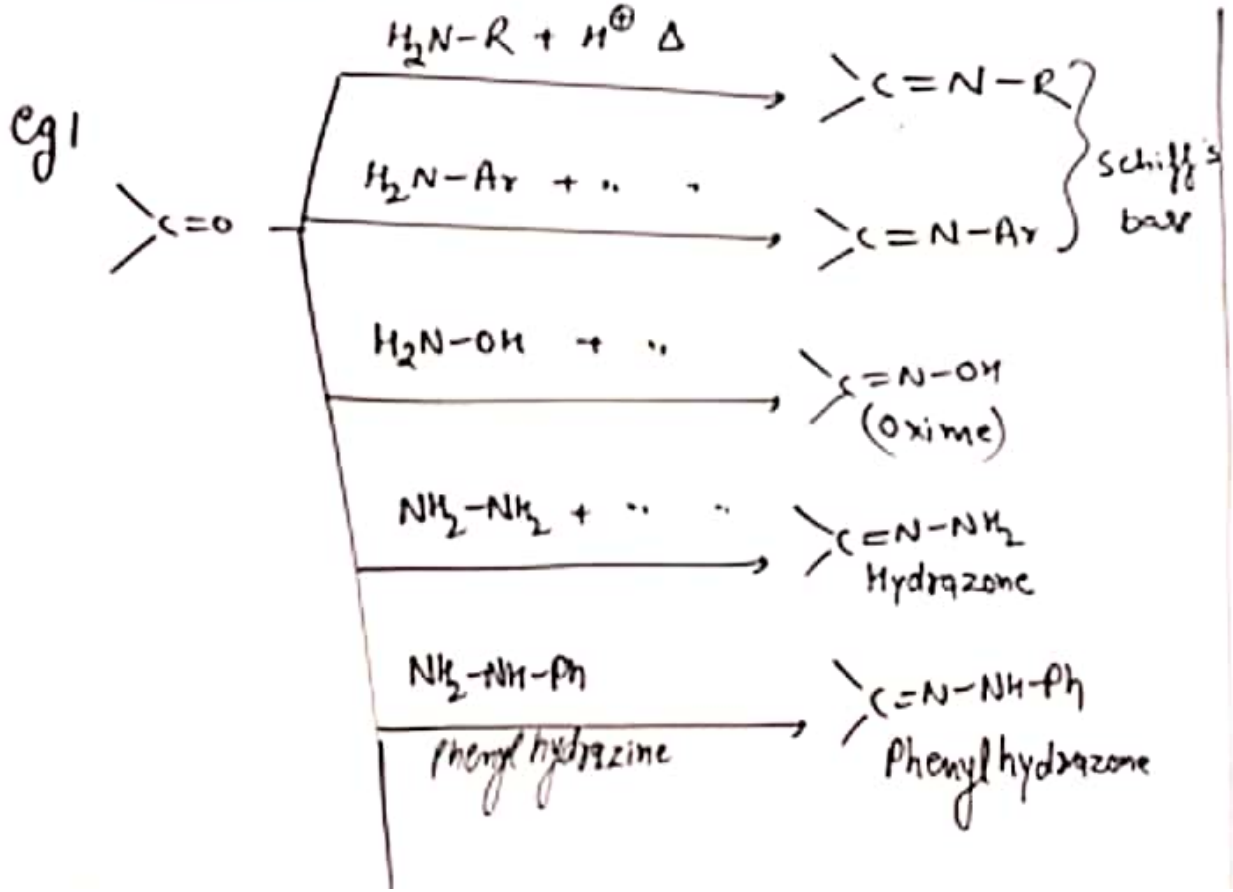
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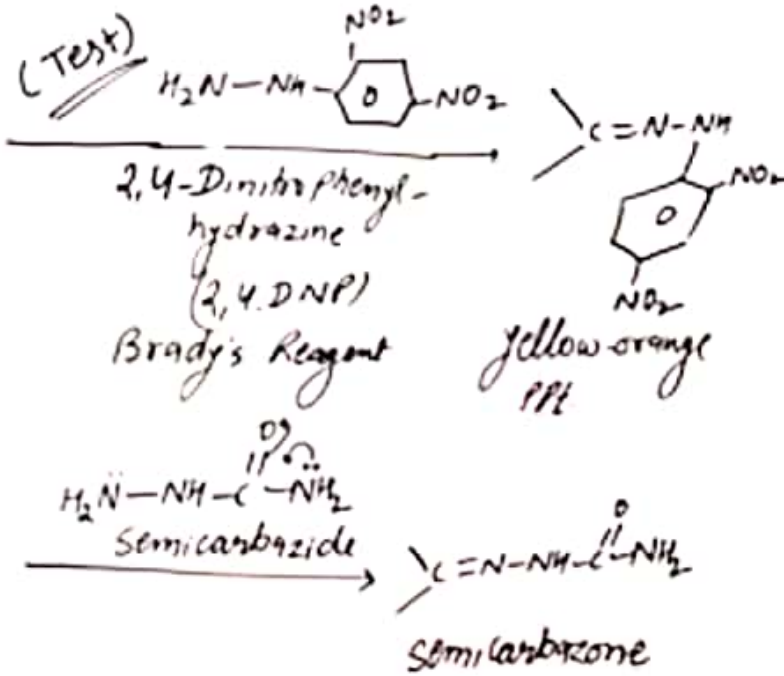
CARBONYL COMPOUNDS

Chemical Reactions:-

[B] Nucleophilic Addition Rxn :-

B8 Reactions with N containing Nucleophile:- (Addition - Elimination)





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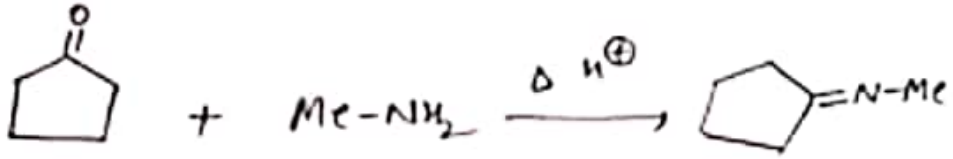
CARBONYL COMPOUNDS

Chemical Reactions:-

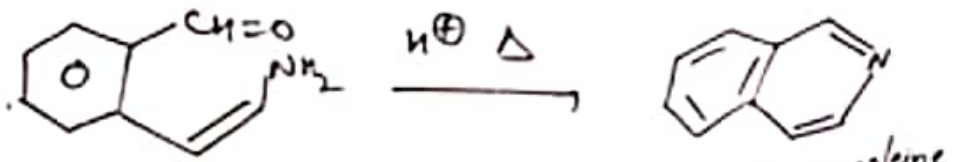
[B] Nucleophilic Addition Rxn :-

B8 Reactions with N containing Nucleophile:- (Addition - Elimination)

eg 2

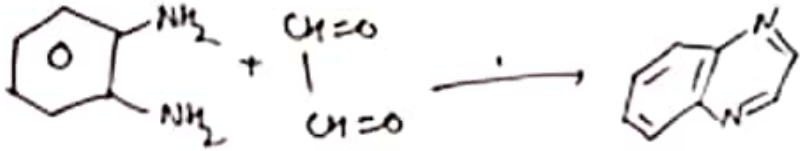


eg 3

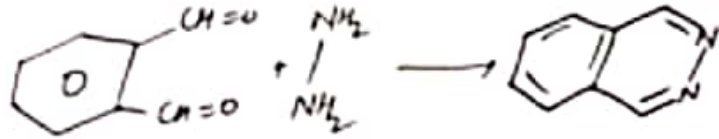


Isoquinolene

eg 4



eg 5



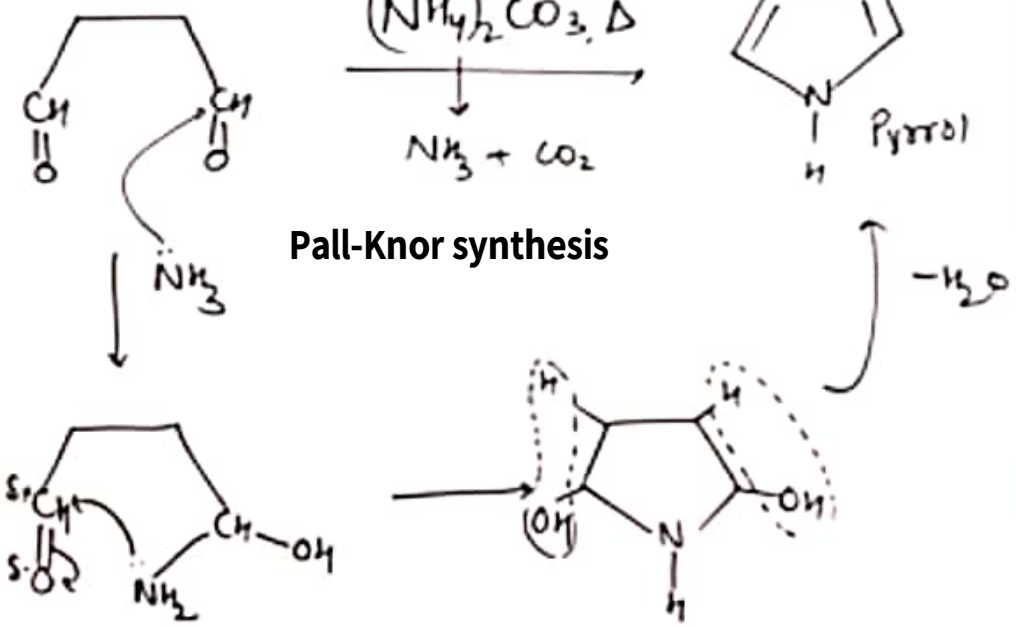
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Chemical Reactions:-

[B] Nucleophilic Addition Rxn:-

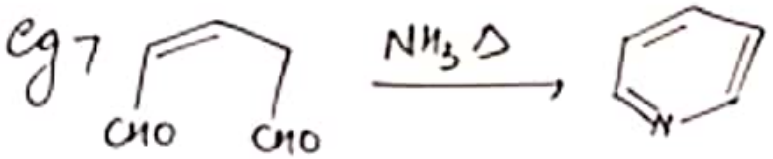
B8 Reactions with N containing Nucleophile:- (Addition - Elimination)

Integ 6



Pall-Knor synthesis

eg 7



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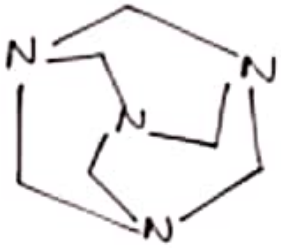
Chemical Reactions:-

[B] Nucleophilic Addition Rxn:-

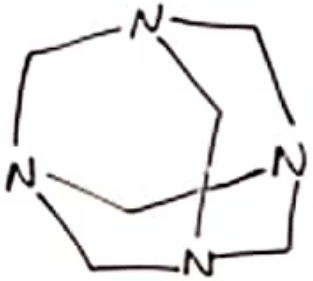
B8 Reactions with N containing Nucleophile:- (Addition - Elimination)

B.8.1 Reaction of ammonia with aldehydes:-

$$\Rightarrow 6\,CH_2{=}O + 4\,NH_3 \xrightarrow[-6H_2O]{\Delta}$$



=

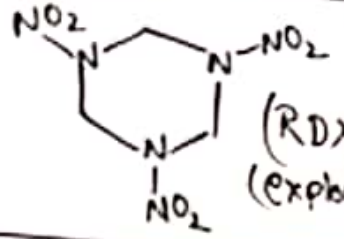


structure similar to P4O6

Urotropine present in urine

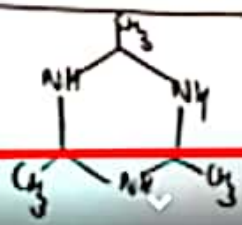
antiseptic

Urotropin $\xrightarrow{\text{conc } HNO_3}$



(RDX) (explosive)

$$\Rightarrow 3\,CH_3{-}CH{=}O + 3NH_3 \longrightarrow$$



Trimethyl triazine

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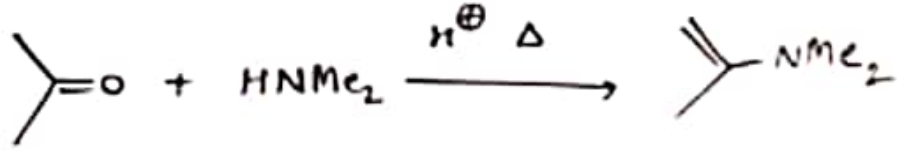
#2|REACTIONS OF ALDEHYDE AND KETONES |NUCLEOPHILIC ADDITION REACTIONS|CARB...

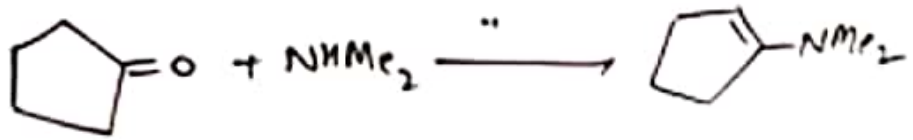
Chemical Reactions:-

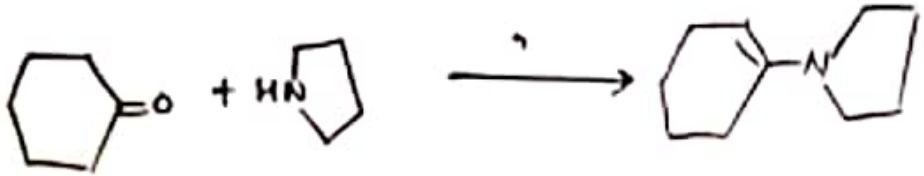
[B] Nucleophilic Addition Rx^n:-

B8 Reactions with N containing Nucleophile:- (Addition - Elimination)

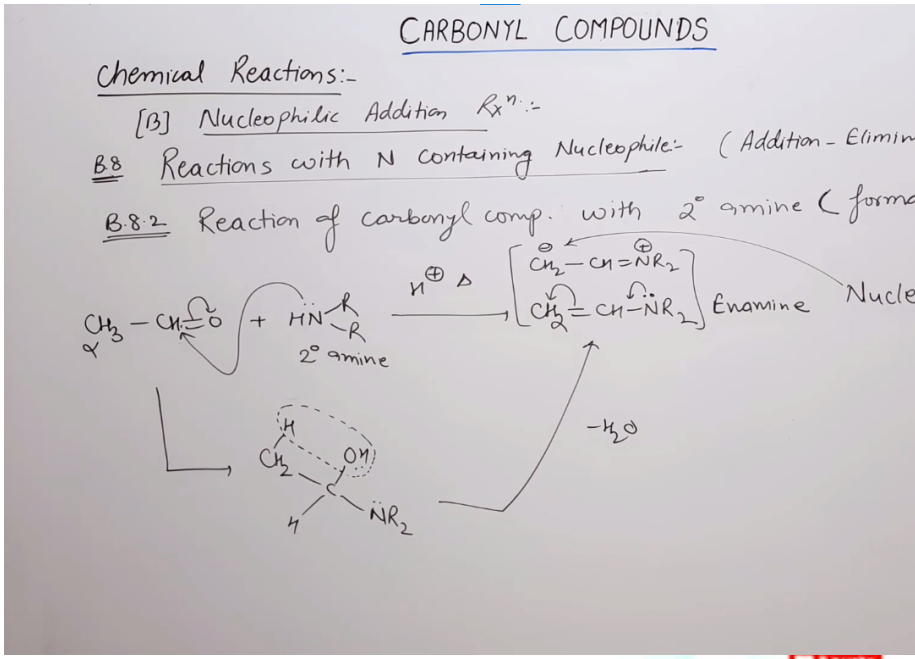
B.8.2 Reaction of carbonyl comp. with 2° amine (formation of Enamine)

Eg 1 

Eg 2 

Eg 3 

pyrrolidine



40:31 / 45:59

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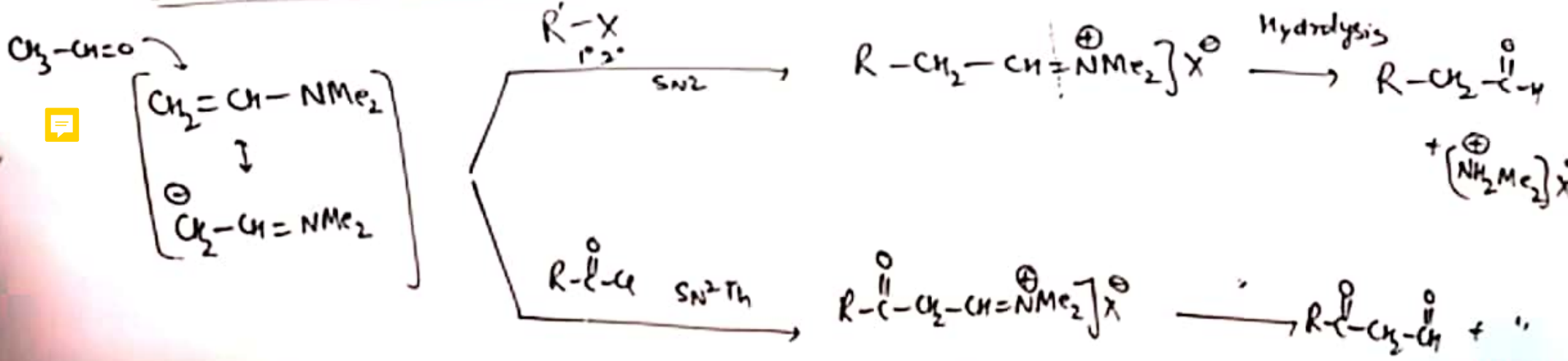
Chemical Reactions:-

[B] Nucleophilic Addition Rx^n:-

B8 Reactions with N containing Nucleophile:- (Addition - Elimination)

B.8.2 Reaction of carbonyl comp. with 2° amine (formation of Enamine)

⇒ Starck - Enamine Reaction:-



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CARBONYL COMPOUNDS

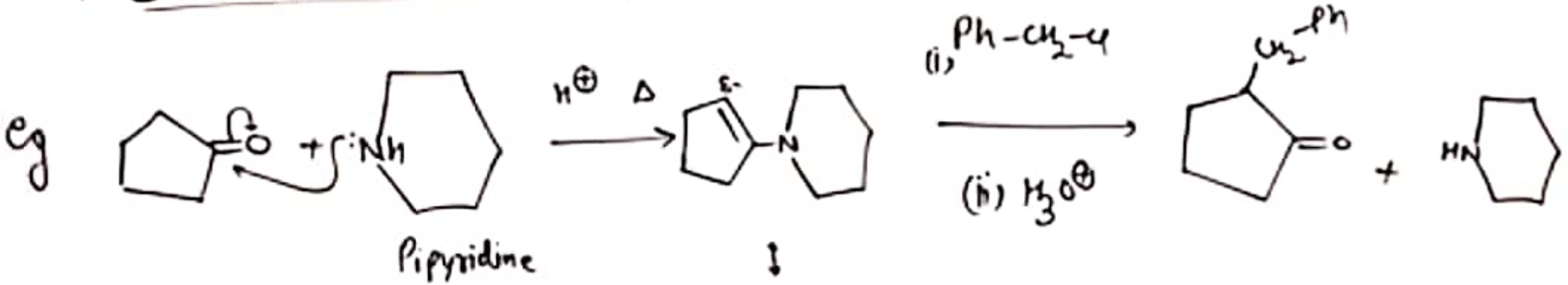
Chemical Reactions:-

[B] Nucleophilic Addition Rxn :-

B8 Reactions with N containing Nucleophile:- (Addition - Elimination)

B.8.2 Reaction of carbonyl comp. with 2° amine (formation of Enamine)

⇒ Starck - Enamine Reaction:-



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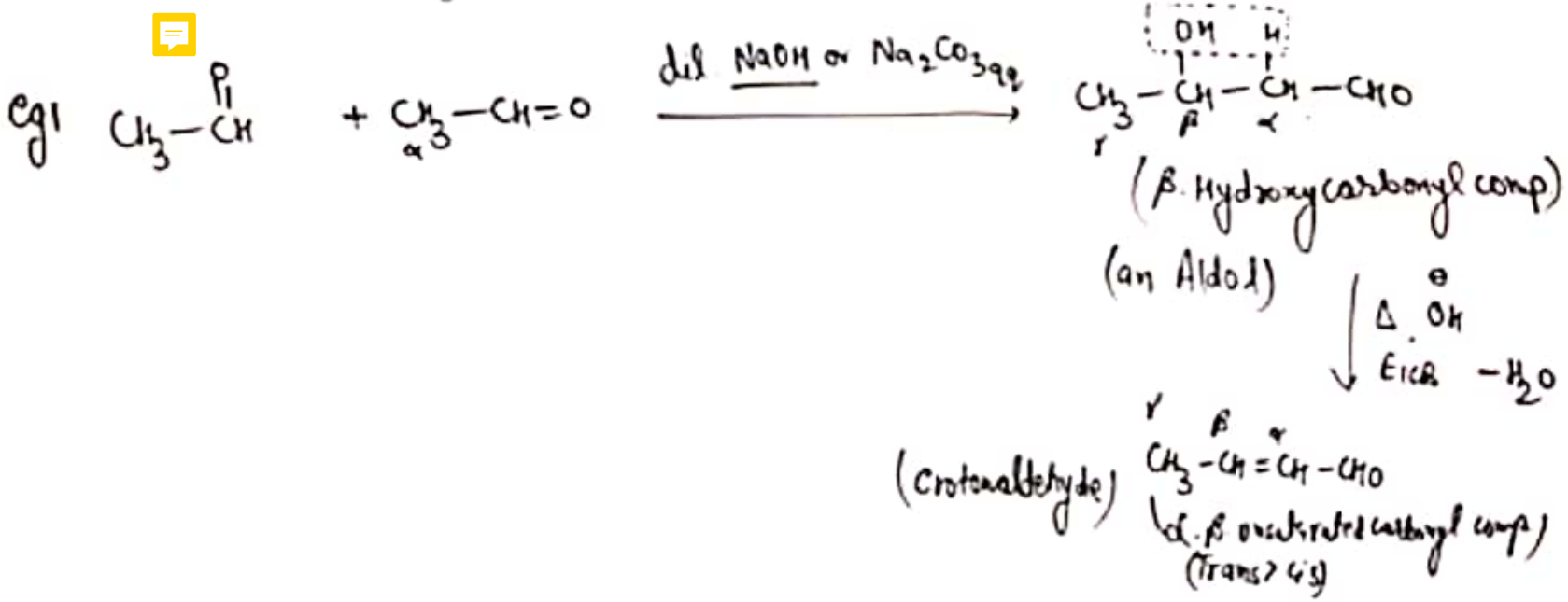
CORBONYL COMPOUNDS

Reactions of carbonyl compounds:-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens:-

C.1 Aldol condensation:- (C_{sp^3})

Reaction of α-acidic hydrogen containing carbonyl compound in presence of dil. base / acid give β Hydroxy carbonyl comp and then α-β unsaturated carbonyl comp.



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CORBONYL COMPOUNDS

Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

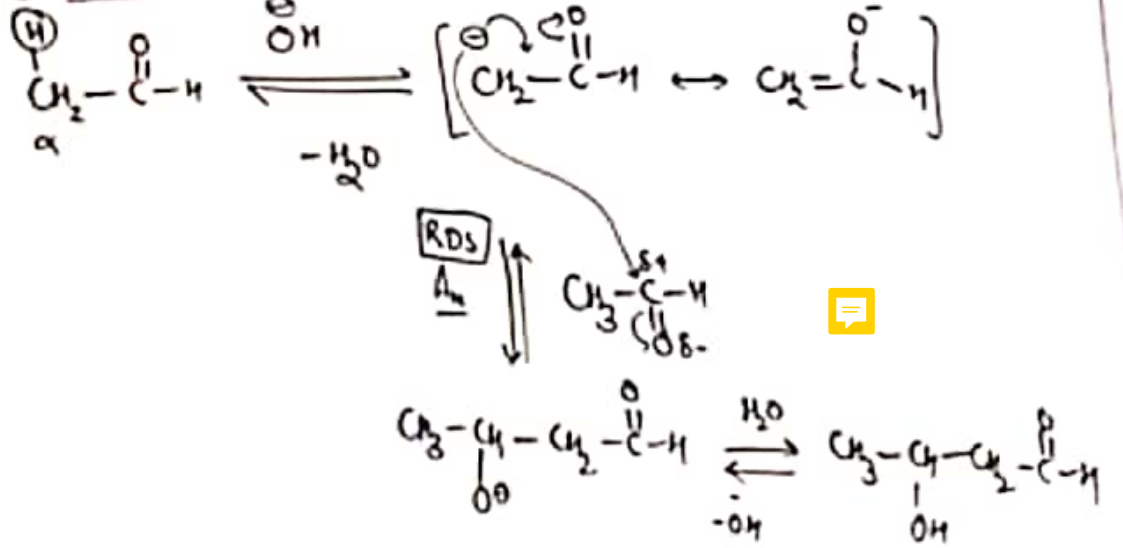
C1 Aldol condensation :-

Mechanism

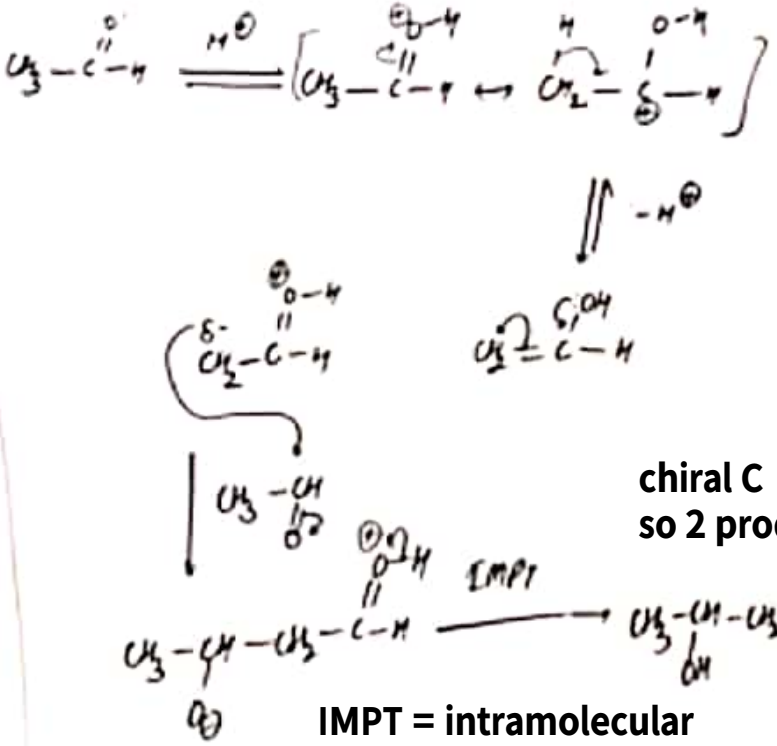
ⓐ In basic medium :



enolate ion



ⓑ In acidic medium :-



chiral C
so 2 products

IMPT = intramolecular proton transfer

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CORBONYL COMPOUNDS

Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

for this condensation to take place, there should be 2 α-H on the same carbon, otherwise rex. stops at the aldol stage

C1 Aldol condensation :-

Eg 2 $CH_3-CH_2-CH=O \xrightarrow{\ominus OH} CH_3-CH_2-\underset{OH}{CH}-\underset{CH_3}{C}-CH=O \xrightarrow{\Delta} CH_3-CH_2-CH=\underset{CH_3}{C}-CH=O$

Eg 3 $Ph-CH_2-CH=O \xrightarrow{\ominus OH} Ph-CH_2-\underset{OH}{CH}-\underset{Ph}{CH}-CHO \xrightarrow{\Delta} Ph-CH_2-CH=\underset{Ph}{C}-CHO$

Eg 4 $Ph-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow{\ominus OH} Ph-\underset{CH_3}{\overset{OH}{\overset{|}{C}}}-CH_2-\overset{O}{\overset{\|}{C}}-Ph \xrightarrow{\Delta} Ph-\underset{CH_3}{C}=CH-\overset{O}{\overset{\|}{C}}-Ph$

Eg 5  $\xrightarrow{\ominus OH,\ \Delta}$ 

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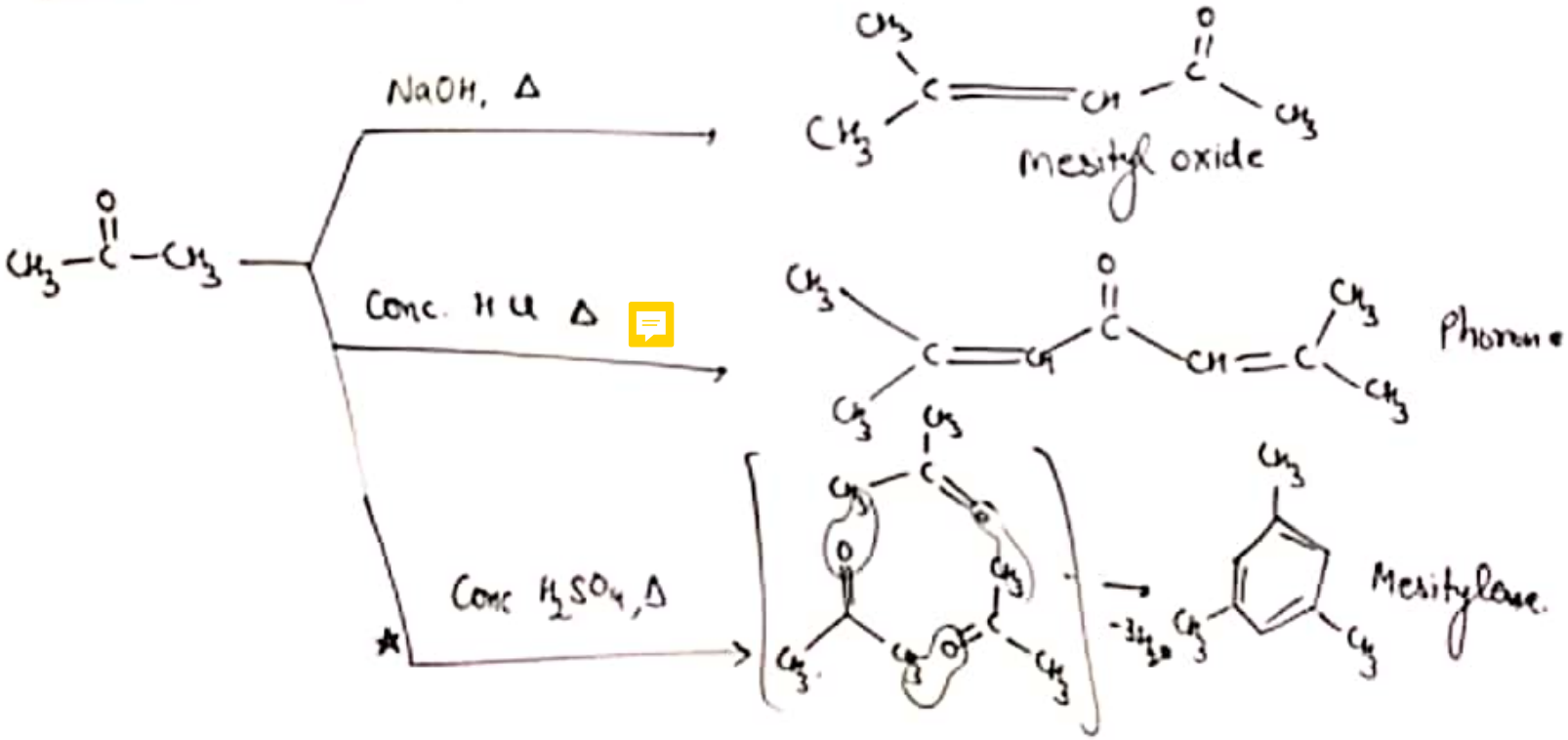
CORBONYL COMPOUNDS

Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

C1 Aldol condensation :-

* eg 6



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CORBONYL COMPOUNDS

Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

C1 Aldol condensation :-

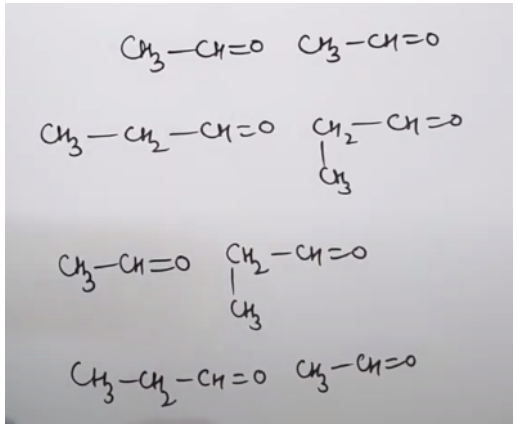
Crossed aldol condensation :- (aldol condensation b/w diff carbonyl comp)

Eg 1 $CH_3-\overset{O}{\overset{\|}{C}}-H$ + $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-H$ $\xrightarrow{\ominus OH,\ \Delta}$

$CH_3-CH=CH-CH=O$

$CH_3-CH_2-CH=\underset{CH_3}{\underset{|}{C}}-CH=O$

Self Products



$CH_3-CH=\underset{CH_3}{\underset{|}{C}}-CH=O$

$CH_3-CH_2-CH=CH-CH=O$

Crossed products

all form all possible Geometric Isomers

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CORBONYL COMPOUNDS

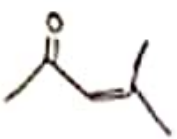
Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

C.1 Aldol condensation :-

Crossed aldol condensation :- (aldol condensation b/w diff carbonyl comp)

eg 8 $Ph-CH=O + CH_3-CH=O \xrightarrow[\text{(Claisen-Schimidt Rxn)}]{\overset{\ominus}{O}H,\ \Delta}$ $Ph-CH=CH-CH=O$ Cinnamaldehyde + $CH_3-CH=CH-CHO$

eg 1 $Ph-CH=O$ +  $\xrightarrow{\overset{\ominus}{O}H}$ $Ph-CH=CH-\overset{O}{\overset{\|}{C}}-CH_3$ + 

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CORBONYL COMPOUNDS

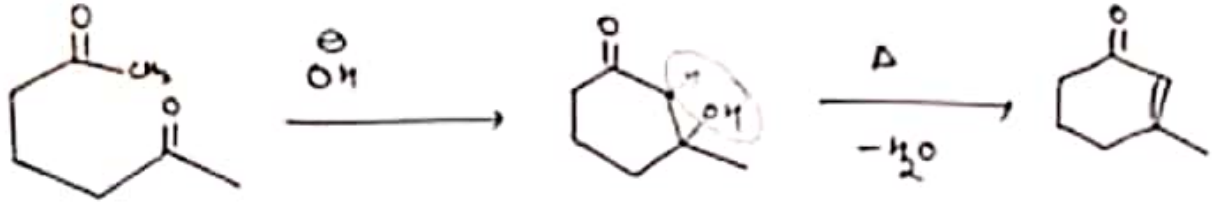
Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

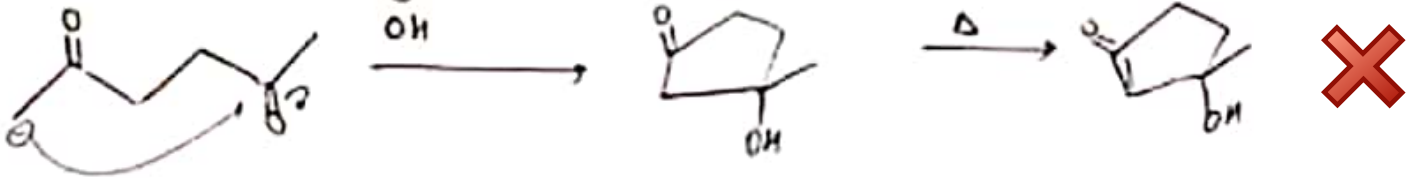
C1 Aldol condensation :-

Intramolecular aldol condensation :-

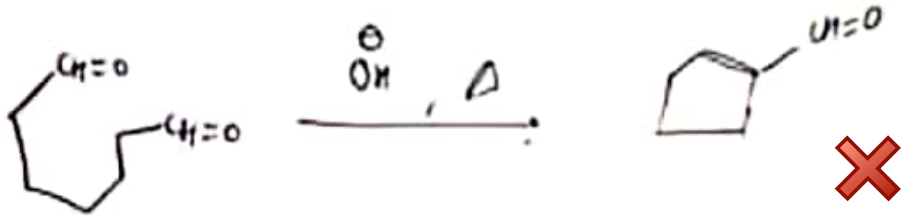
Eg 10



Eg 11



Eg 12



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CORBONYL COMPOUNDS

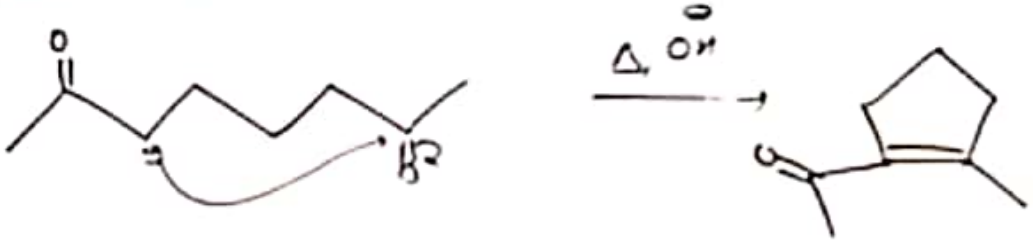
Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

C1 Aldol condensation :-

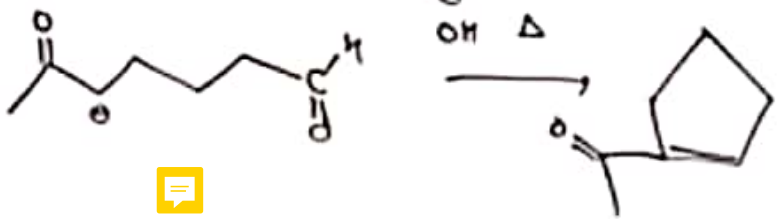
Intramolecular aldol condensation :-

Eg 13



- 3 < 5 > 7
- 4 < 6 > 8

Eg 14



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CORBONYL COMPOUNDS

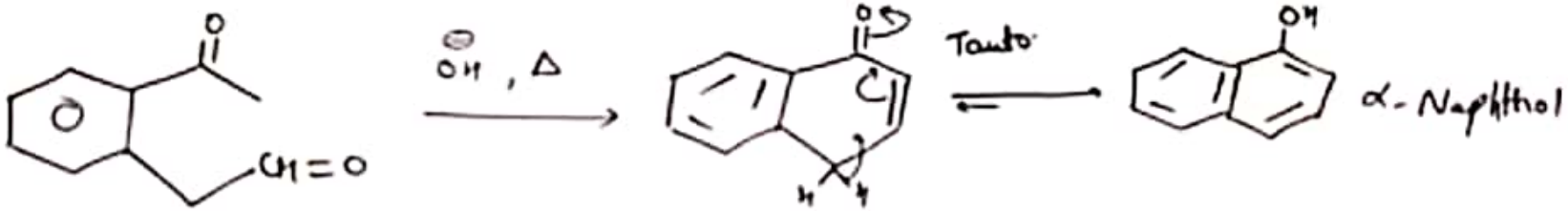
Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

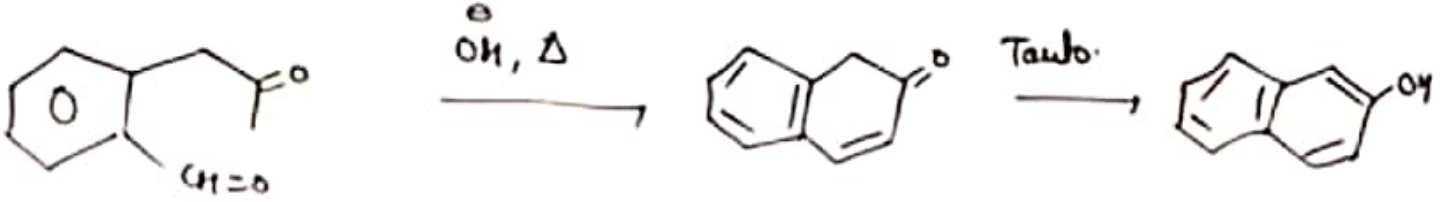
C1 Aldol condensation :-

Intramolecular aldol condensation :-

Eg 15



Eg 16



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CORBONYL COMPOUNDS

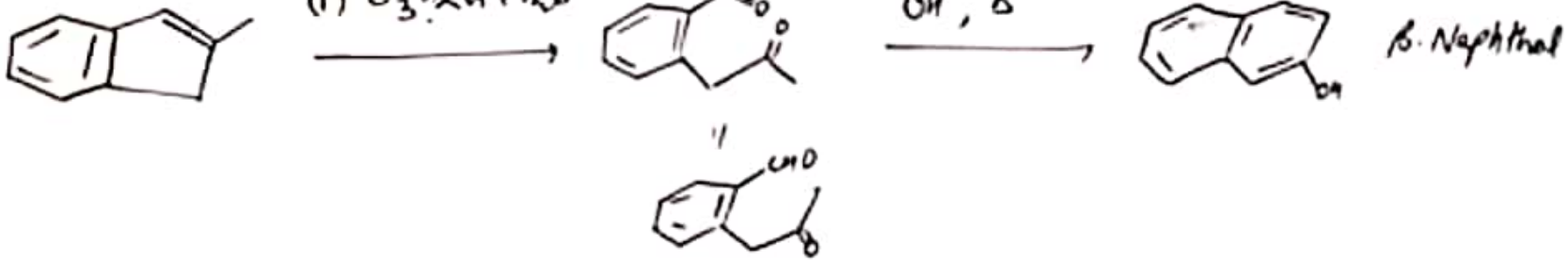
Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

C1 Aldol condensation :-

Intramolecular aldol condensation :-

eg 17



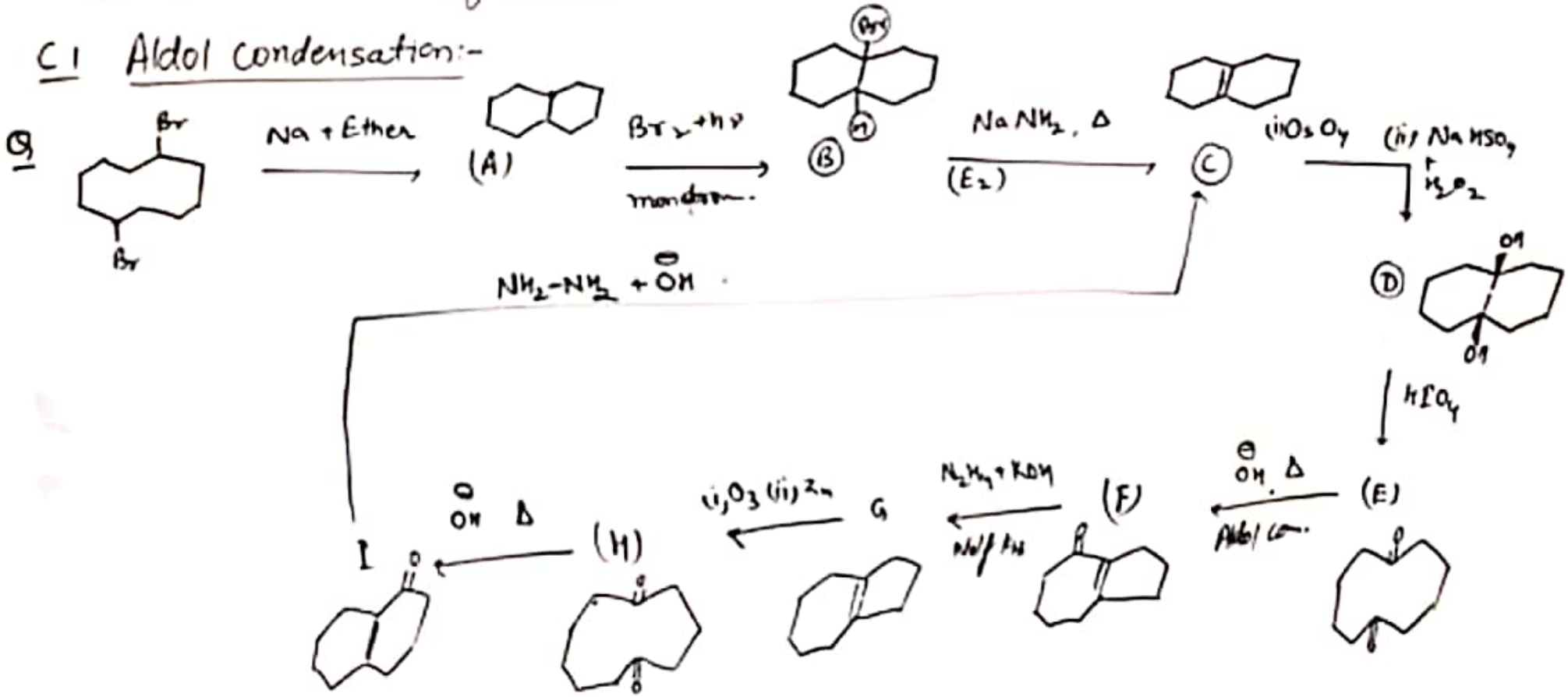
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CORBONYL COMPOUNDS

Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

C1 Aldol condensation :-



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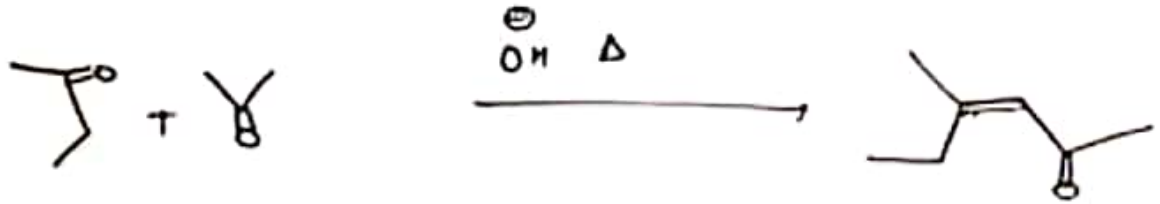
CORBONYL COMPOUNDS

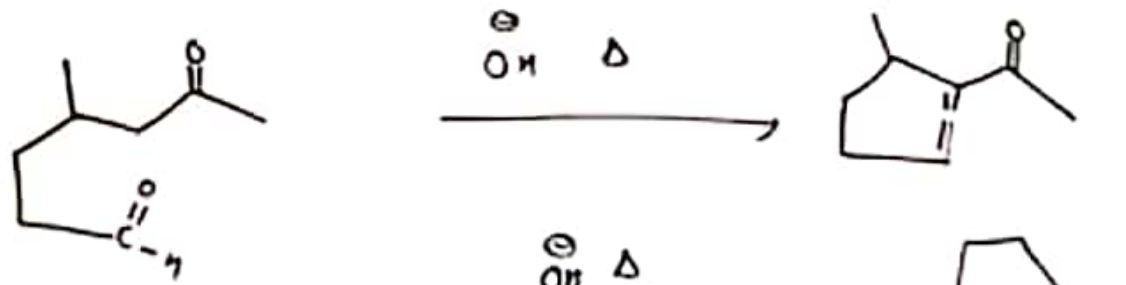
Reactions of carbonyl compounds -

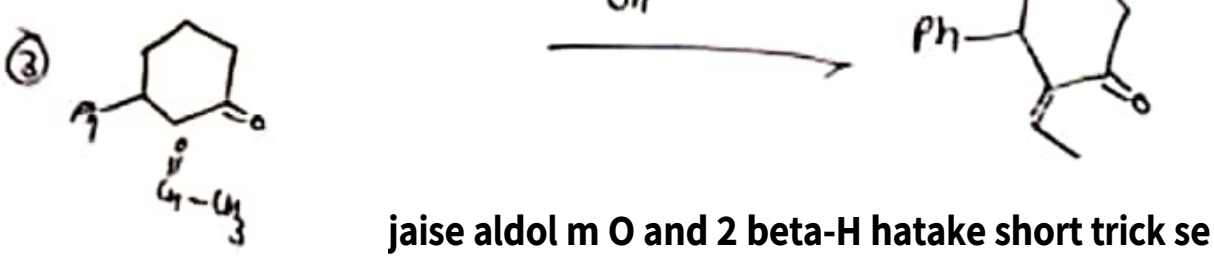
[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

C1 Aldol condensation :-

Q Give suitable structures of reactants for following aldol conden reactions.

① 

② 

③ 

jaise aldol m O and 2 beta-H hatake short trick se product banaya tha, waise hi uska reverse process yaha kr do

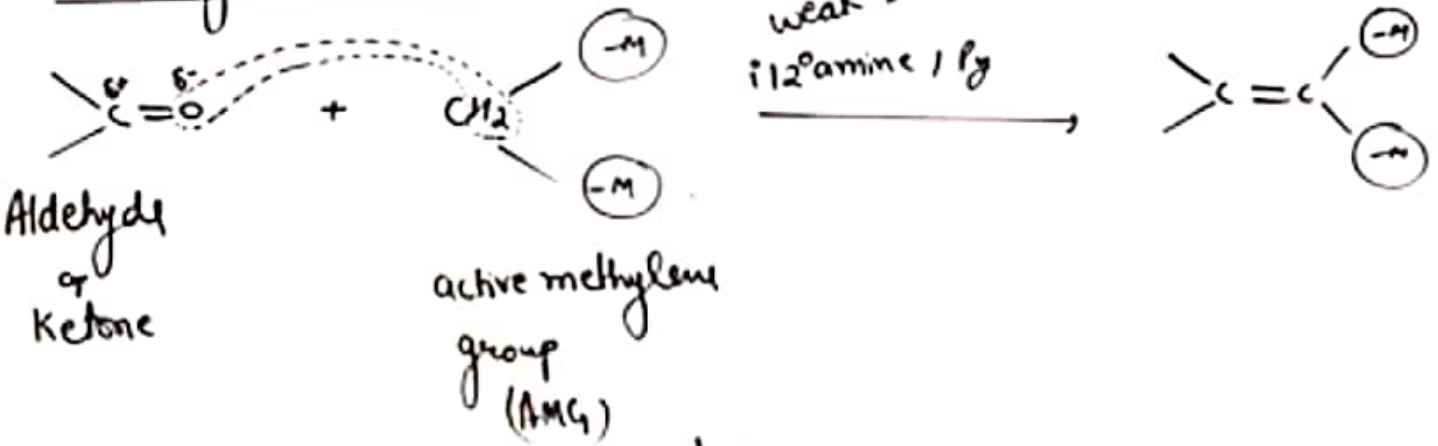
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CORBONYL COMPOUNDS

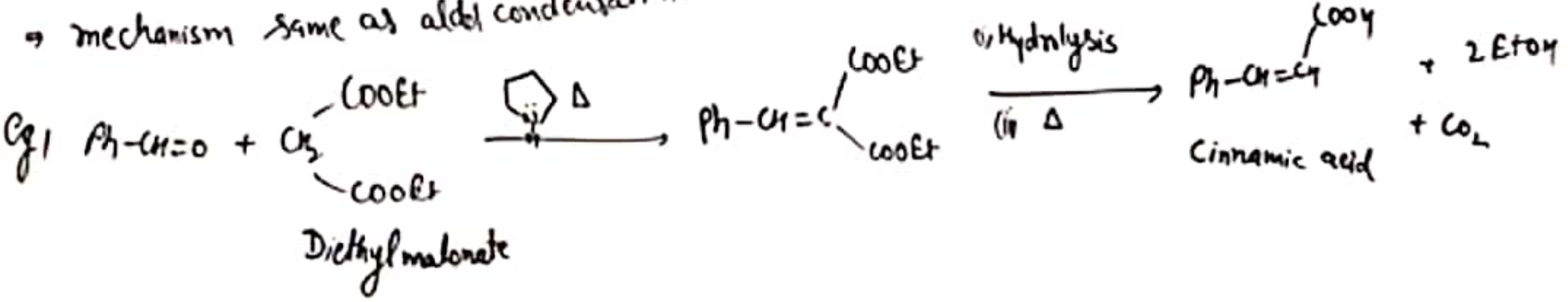
Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

C.2 Knovenagel Reaction :-



⇒ mechanism same as aldol condensation :-



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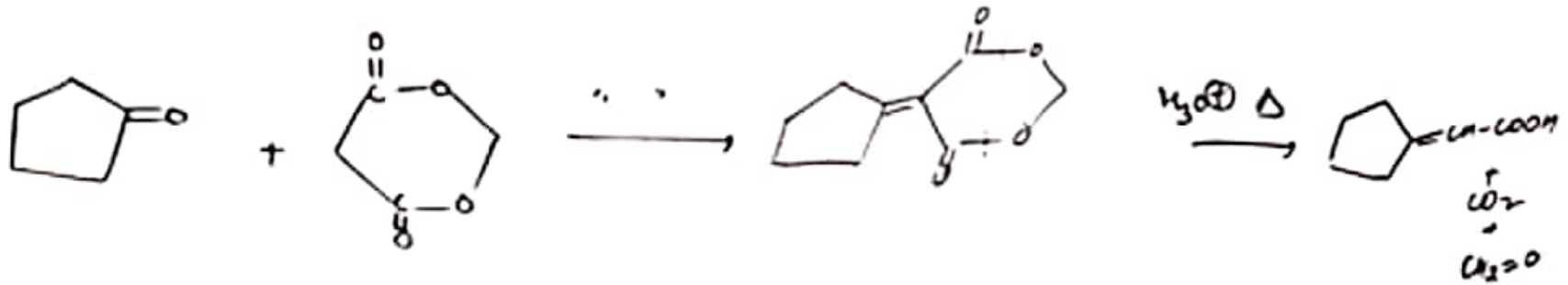
CORBONYL COMPOUNDS

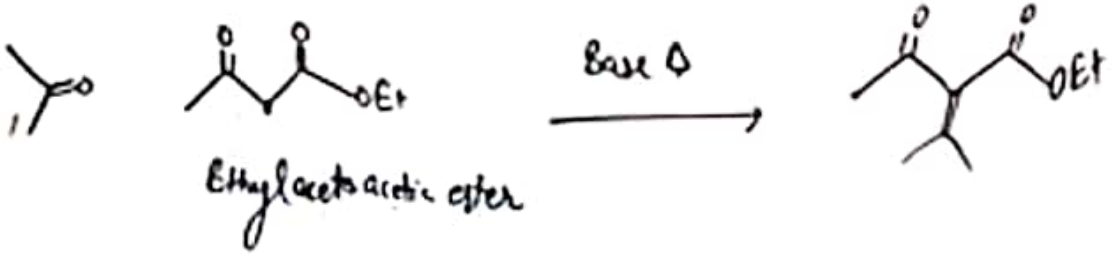
Reactions of carbonyl compounds:-

[C] Reactions involving enolate ions / Reactions of α-acidic hydrogens:-

C.2 Knovenagel Reaction:-

eg 2 $CH_3-CH=O + CH_3-NO_2 \xrightarrow{Py,\ \Delta} CH_3-CH=CH-NO_2$

eg 3 

eg 4 

Ethylacetoacetic ester

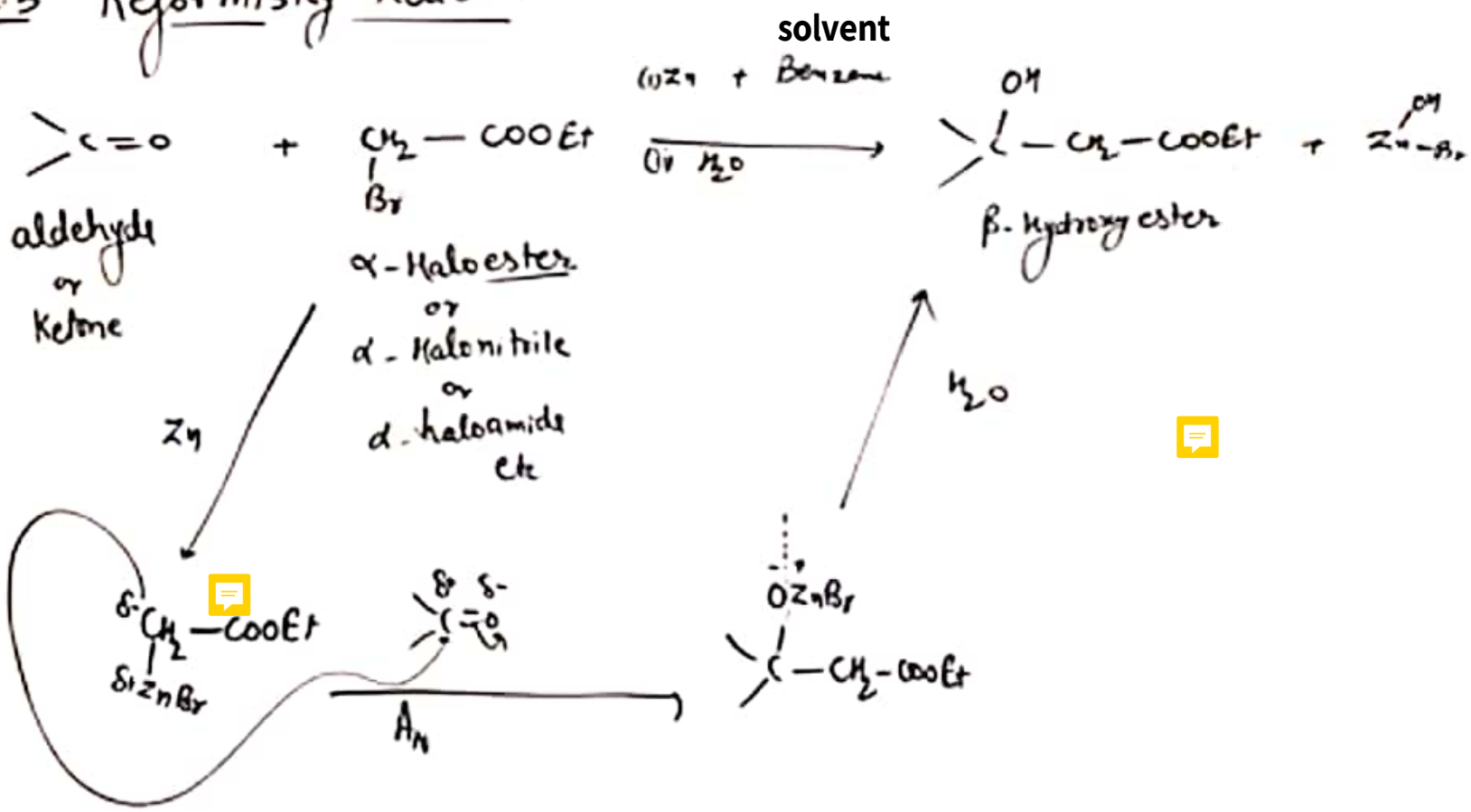
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CORBONYL COMPOUNDS

Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

C.3 Reformtsky Reaction :-



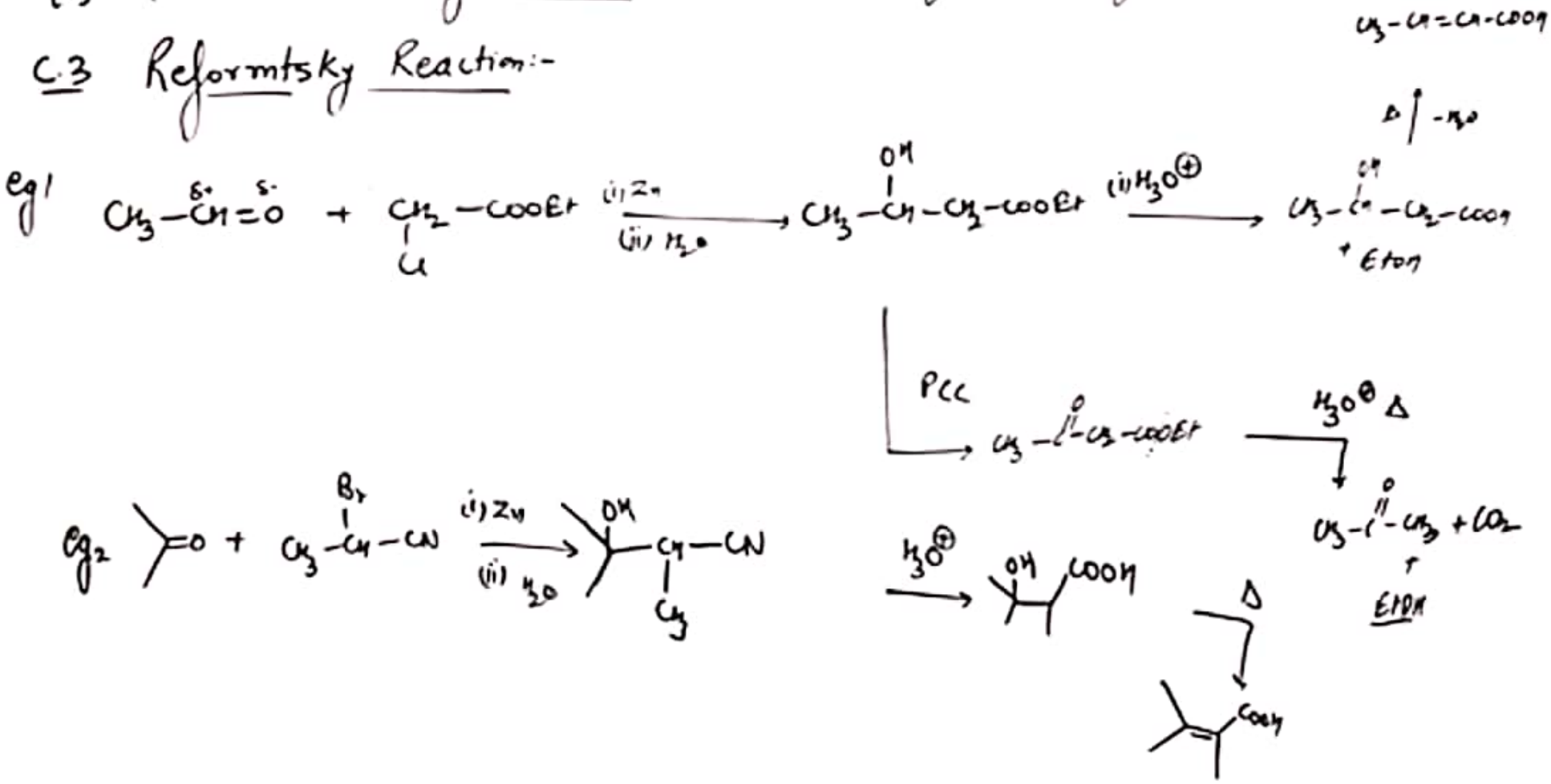
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CORBONYL COMPOUNDS

Reactions of carbonyl compounds:-

[C] Reactions involving enolate ions / Reactions of α-acidic hydrogens:-

C.3 Reformtsky Reaction:-



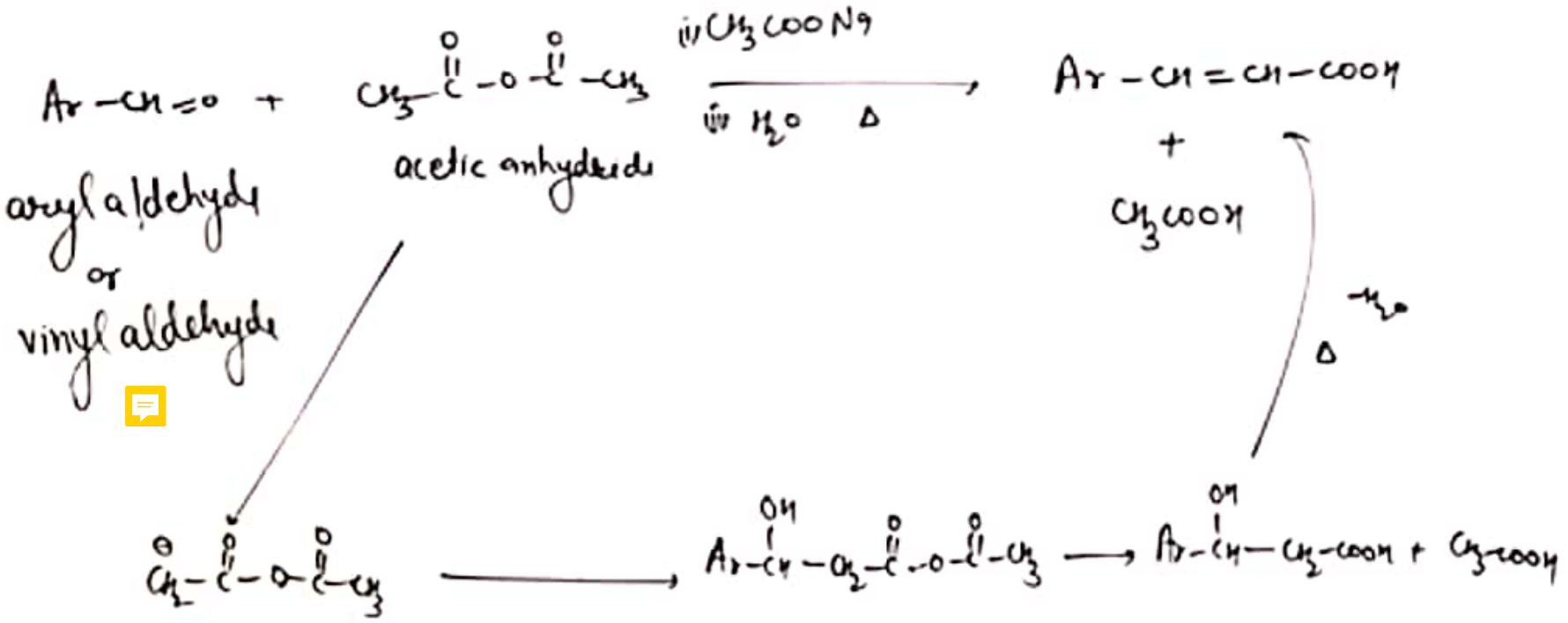
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CORBONYL COMPOUNDS

Reactions of carbonyl compounds :-

[C] Reactions involving enolate ions / Reactions of α acidic hydrogens :-

C4 Perkins Reaction :-



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CORBONYL COMPOUNDS

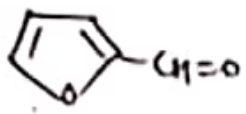
Reactions of carbonyl compounds:-

[C] Reactions involving enolate ions / Reactions of α-acidic hydrogens:-

C4 Perkins Reaction:-

Eg 1 $Ph-CHO + Ac_2O \xrightarrow{AcONa} Ph-CH=CH-COOH + AcOH$
(cis < Trans)

Eg 2 $CH_2=CH-CHO + Ac_2O \xrightarrow{"} CH_2=CH-CH=CH-COOH + "$

Eg 3  $+ Ac_2O \xrightarrow{"}$  $CH=CH-COOH + "$

Furfural

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CARBONYL COMPOUNDS

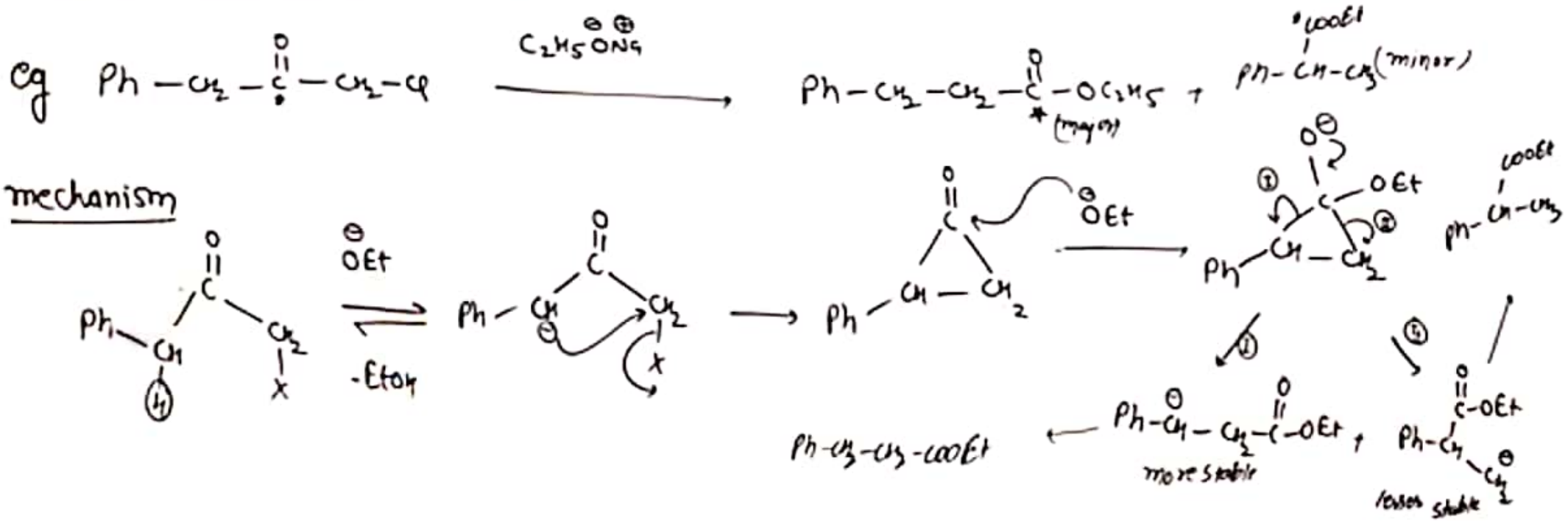
CHEMICAL REACTIONS :-

[c] Reactions involving enolate ions / α-acidic hydrogen containing compounds :-

C.5 FAVORSKII REARRANGEMENT :-

Reaction of α-Haloketones with strong base and form ester/acid

- intermediate = cyclopropanone -



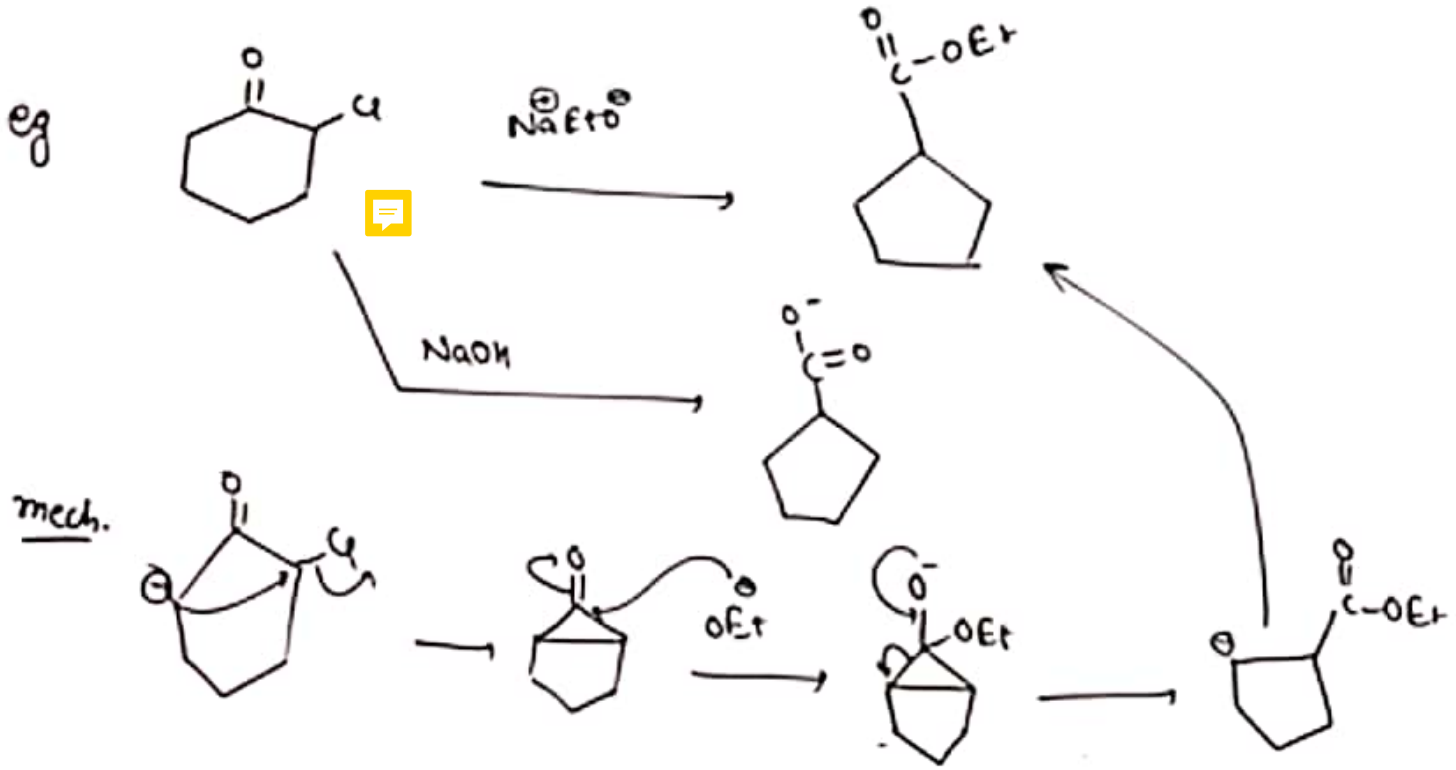
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CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[c] Reactions involving enolate ions / α-acidic hydrogen containing compounds :-

C.5 FAVORSKII REARRANGEMENT :-



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CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[c] Reactions involving enolate ions / α. acidic hydrogen containing compounds :-

C6 CLAISEN CONDENSATION :-

Reaction of α - hydrogen containing ester in presence of strong base and form β. Keto ester. (S_N^2 m)

Eg: $2\,CH_3-\overset{O}{\overset{\|}{C}}-O-Et \xrightarrow{EtO^{\ominus} \text{ or } NH_2^{\ominus}} CH_3-\overset{O}{\overset{\|}{C}}-\underset{\alpha}{CH_2}-\overset{O}{\overset{\|}{C}}-O-C_2H_5$

Ethylacetate

Ethyl acetyl acetate (β Keto ester)

or

Ethyl aceto acetic ester

Mechanism

$CH_3-\overset{O}{\overset{\|}{C}}-\underset{H}{\underset{|}{CH_2}}-\overset{O}{\overset{\|}{C}}-OEt \rightleftharpoons CH_2^{\ominus}-\overset{O}{\overset{\|}{C}}-OEt + EtOH$ ($-EtOH$)

$CH_2^{\ominus}-\overset{O}{\overset{\|}{C}}-OEt + CH_3-\overset{O}{\overset{\|}{C}}-OEt \xrightarrow{(RDS)(S_N2)} CH_3-\underset{OEt}{\overset{O^{\ominus}}{\overset{|}{C}}}-CH_2-\overset{O}{\overset{\|}{C}}-OEt \xrightarrow{-EtO^{\ominus}} CH_3-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-OEt$

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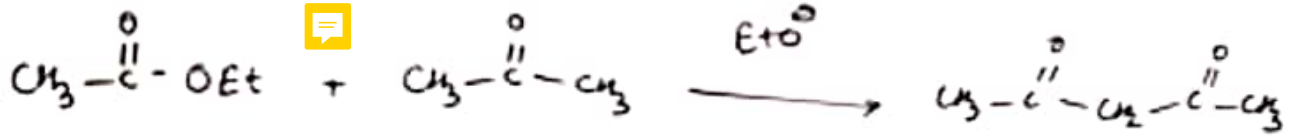
CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[c] Reactions involving enolate ions / α-acidic hydrogen containing compounds :-

C6 CLAISEN CONDENSATION :-

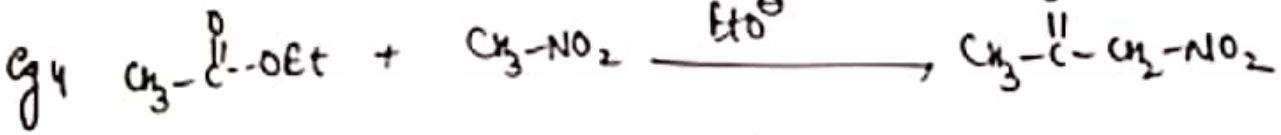
eg 2



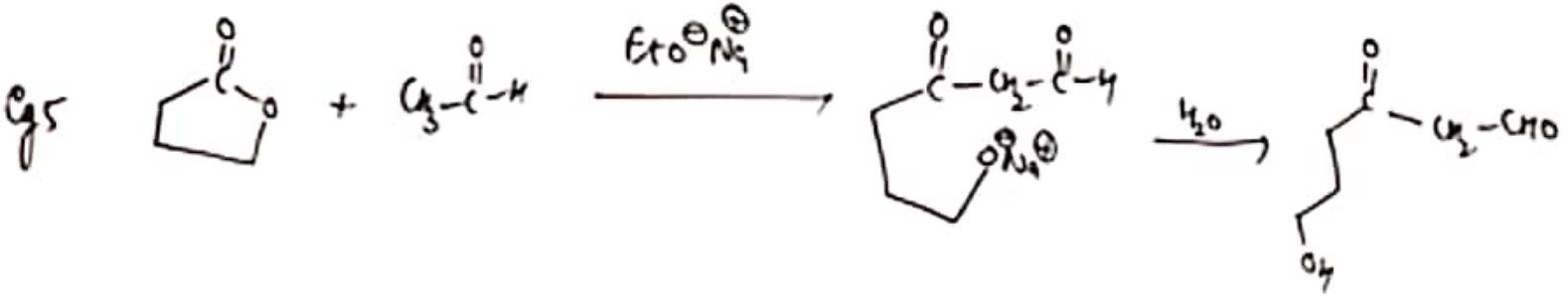
eg 3



eg 4



eg 5



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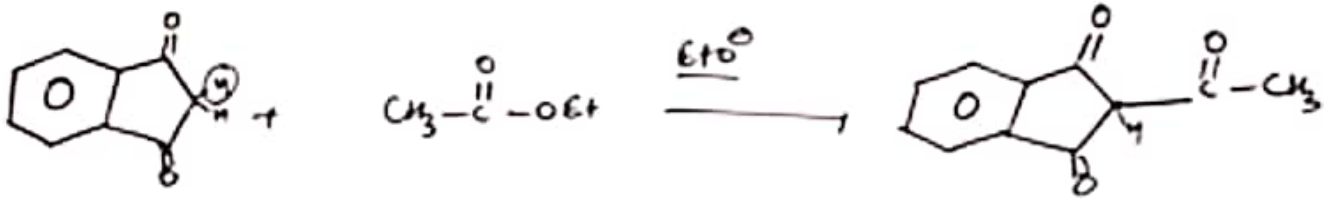
CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

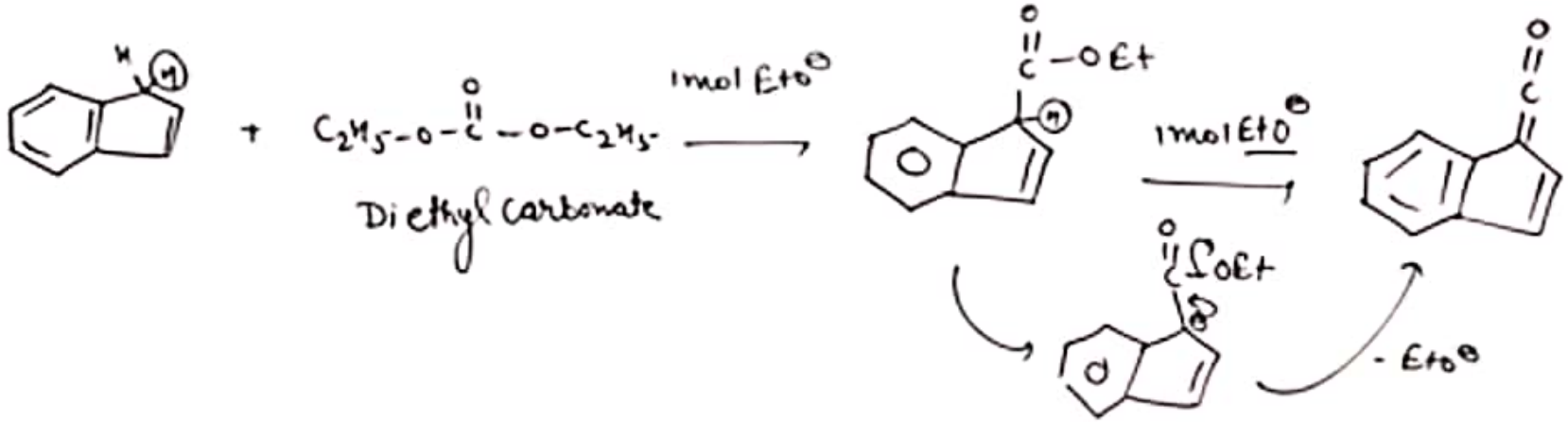
[c] Reactions involving enolate ions / α-acidic hydrogen containing compounds :-

C6 CLAISEN CONDENSATION :-

eg 6



eg 7



Scanned with CamScanner

CARBONYL COMPOUNDS

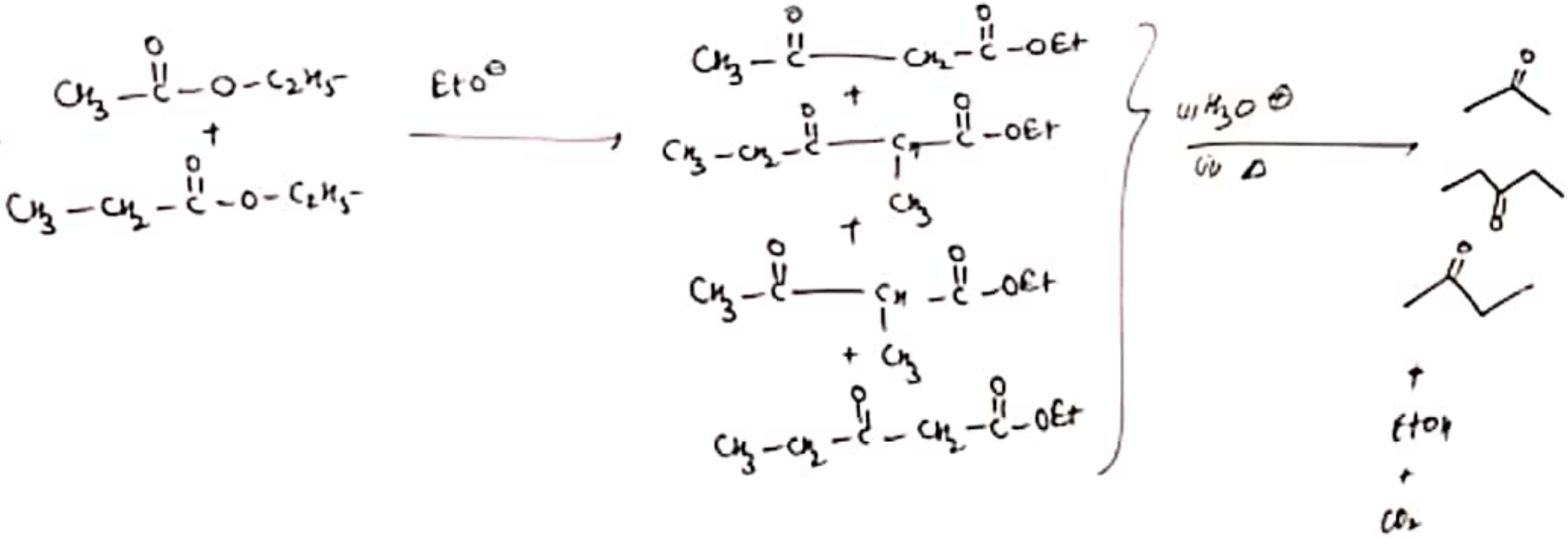
CHEMICAL REACTIONS :-

[c] Reactions involving enolate ions / α-acidic hydrogen containing compounds :-

C6 CLAISEN CONDENSATION :-

⇒ Crossed claisen condensation :-

Eg :-



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CARBONYL COMPOUNDS

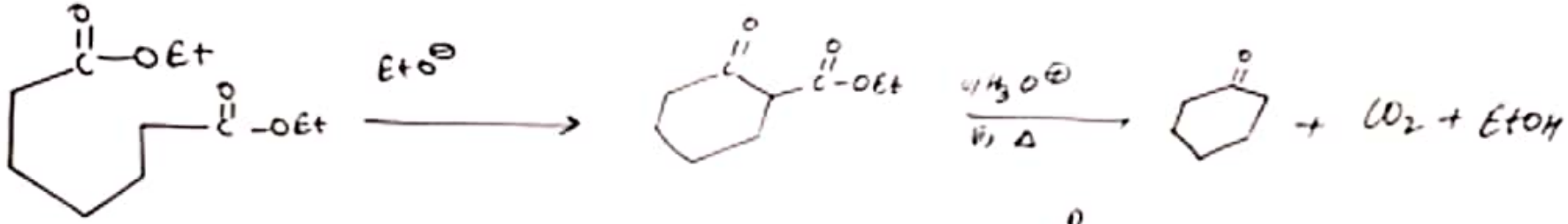
CHEMICAL REACTIONS :-

[c] Reactions involving enolate ions / α-acidic hydrogen containing compounds :-

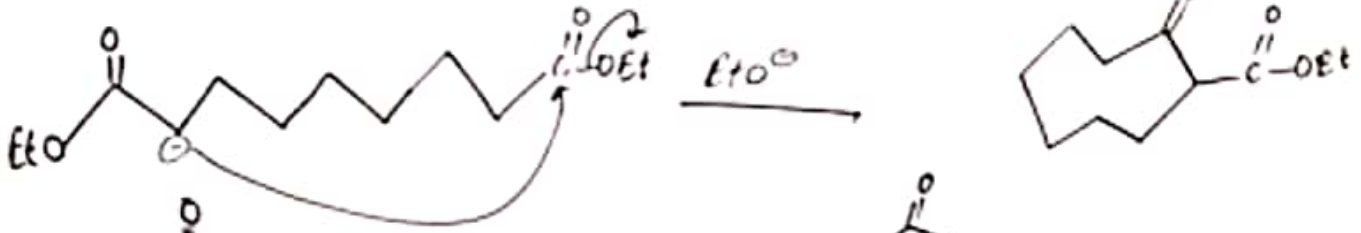
C6 CLAISEN CONDENSATION :-

* Intramolecular Claisen condensation (DIECKMANN'S REACTION)

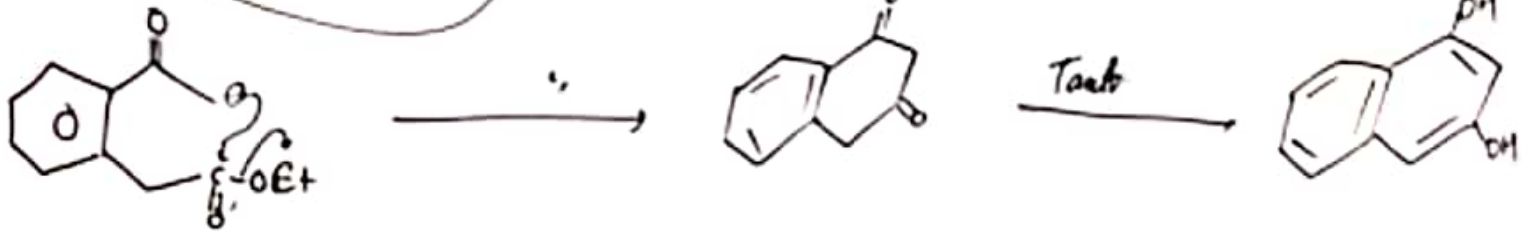
eg 9



eg 10



eg 11



CONTINUE FROM HERE FROM SCREENSHOTS

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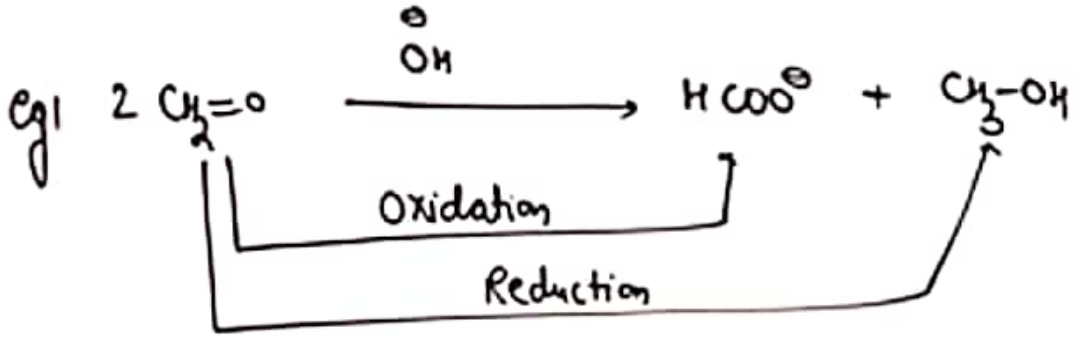
CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

[D1] Cannizzaro Reaction :-

This reaction is disproportional reaction of aldehyde (having no α hydrogens like $CH_2=O$,  , $Ph-CHO$ etc) in presence of strong base and form 1° alcohol and salt of carboxylic acid.



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CARBONYL COMPOUNDS

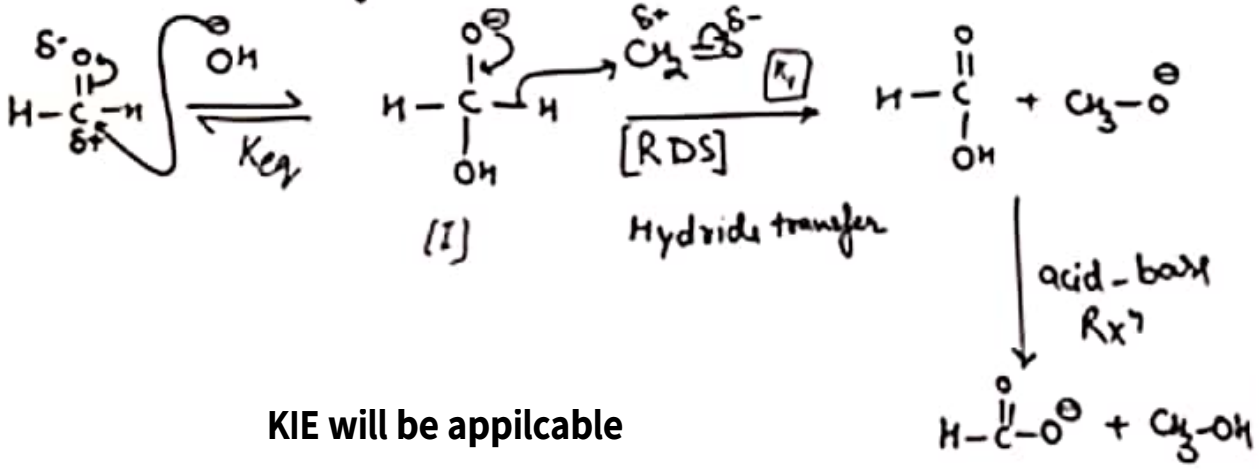
CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

[D1] Cannizzaro Reaction :-

Mechanism

⑨ In low conc. of base :-



KIE will be appilcable

→ 3rd order Rxn which is pretty rare

- Rate $\propto [CH_2=O]^2 [\bar{O}H]^1$

- Rate of Rxn $= K_1 [I] [CH_2=O]$

$$K_{eq} = \frac{[I]}{[\bar{O}H][CH_2=O]}$$

$$[I] = K_{eq}[\bar{O}H][CH_2=O]$$

Rate of Rxn $= K_1 K_{eq} [\bar{O}H][CH_2=O][CH_2=O]$

$= K [CH_2=O]^2 [\bar{O}H]$

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CARBONYL COMPOUNDS

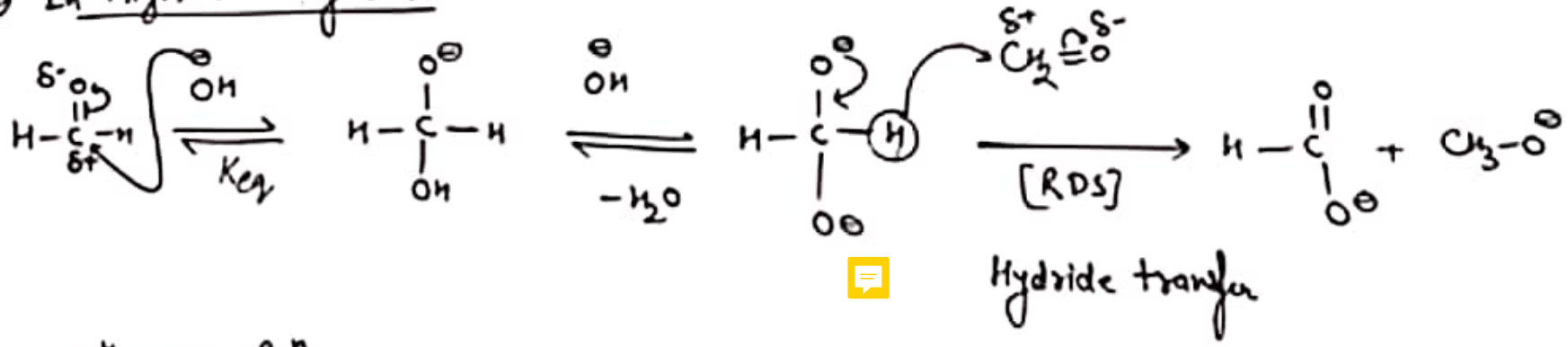
CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

[D1] Cannizzaro Reaction :-

Mechanism

(a) In high conc. of base :-



- 4th order Rxn which is even more rare
- Rate of Rxn $= K[CH_2=O]^2[\bar{O}H]^2$

In Cannizzaro reaction KIE is applicable in both mechanisms

- Reactivity $\boxed{CH_2=O > CD_2=O}$

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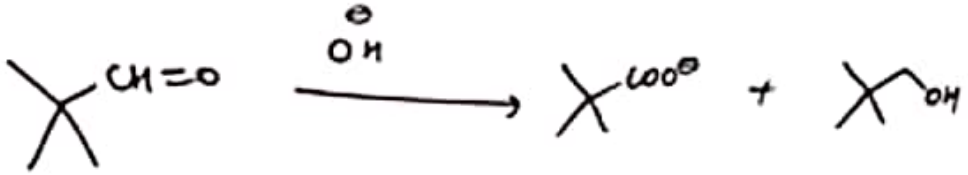
CARBONYL COMPOUNDS

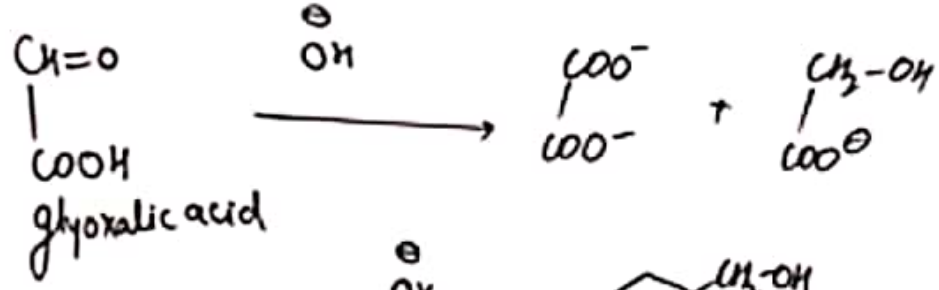
CHEMICAL REACTIONS :-

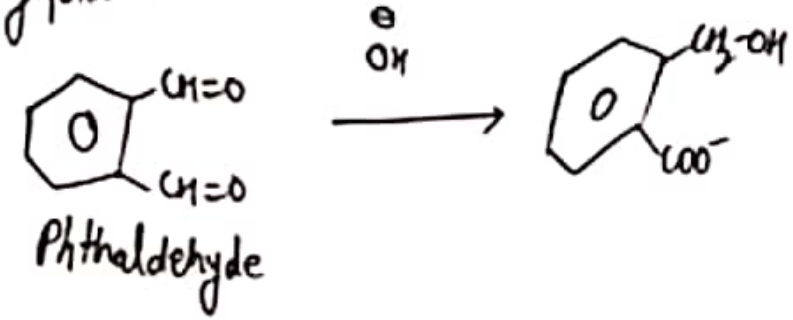
[D] OTHER REACTIONS :-

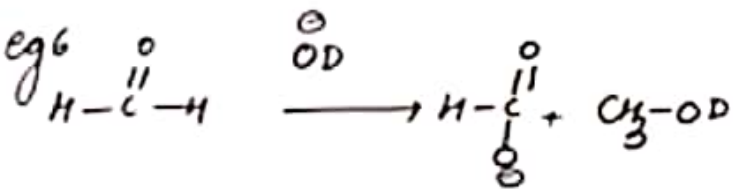
[D1] Cannizzaro Reaction :-

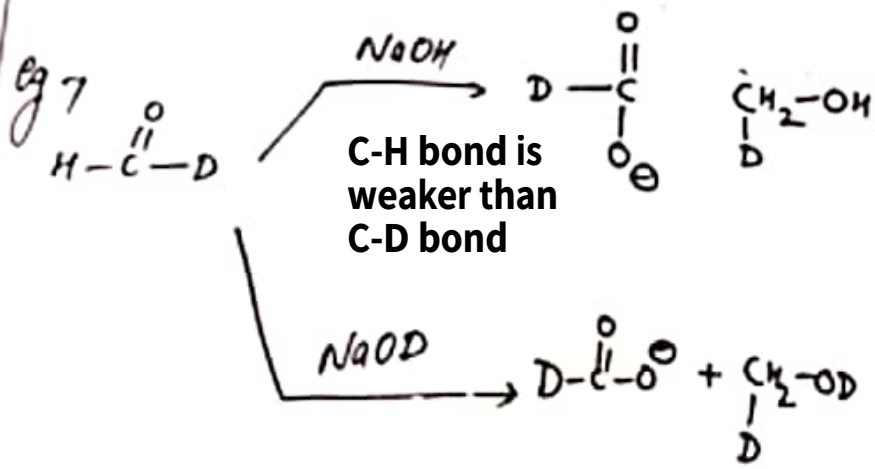
eg 2 $Ph-CHO \xrightarrow{\ominus OH} Ph-COO^- + Ph-CH_2-OH$

eg 3 

eg 4 

eg 5 

eg 6 

eg 7 

C-H bond is weaker than C-D bond

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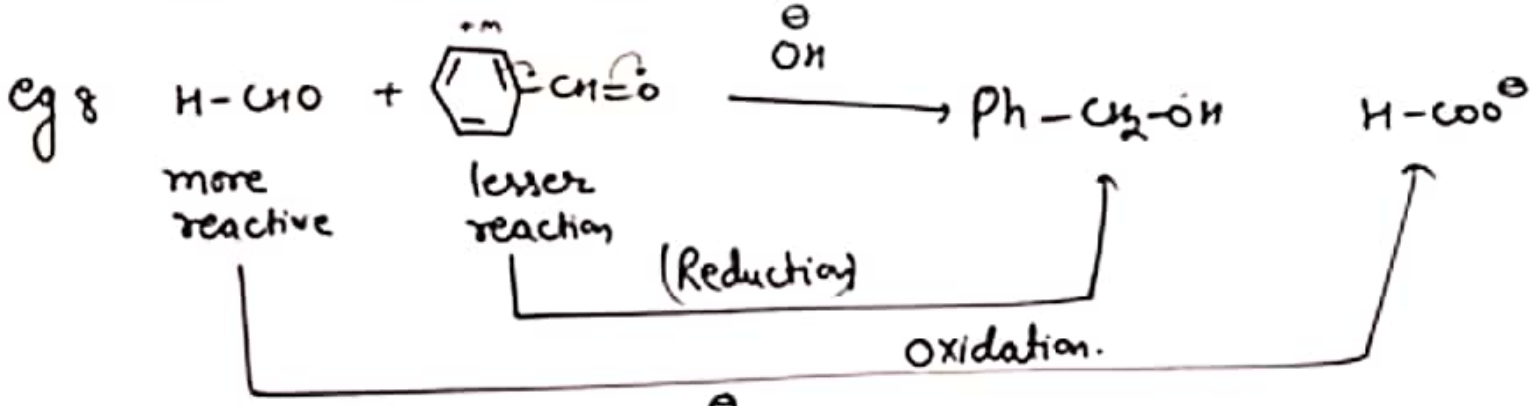
CARBONYL COMPOUNDS

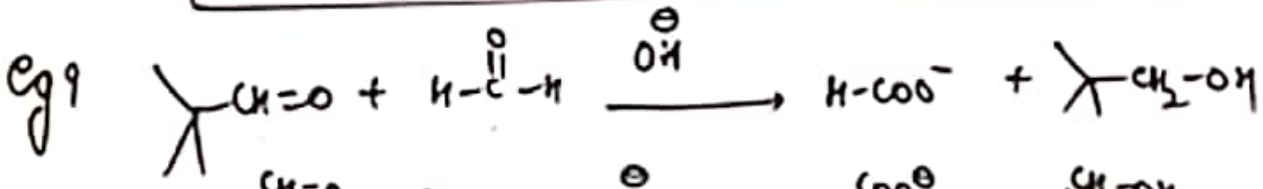
CHEMICAL REACTIONS :-

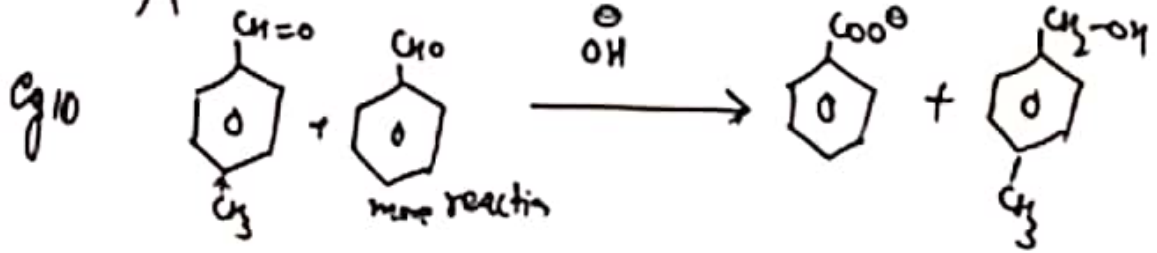
[D] OTHER REACTIONS :-

[D.1] Cannizzaro Reaction :-

crossed Cannizzaro Rxn







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CARBONYL COMPOUNDS

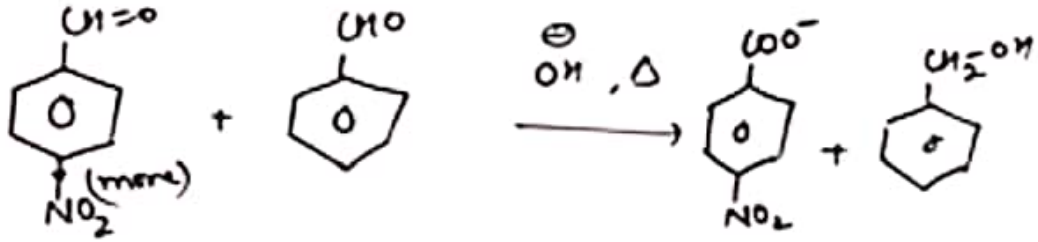
CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

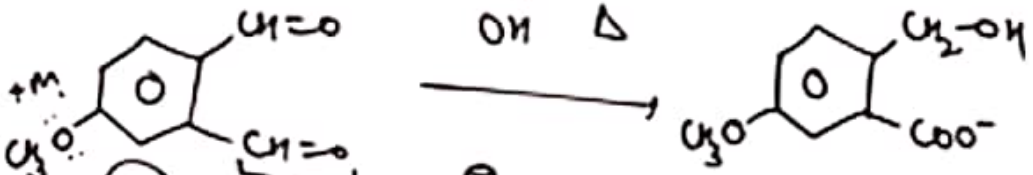
[D.1] Cannizzaro Reaction :-

crossed Cannizzaro Rx^n

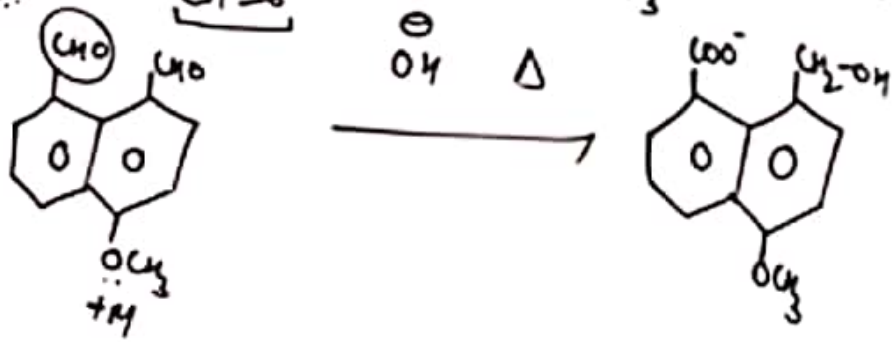
eg 11



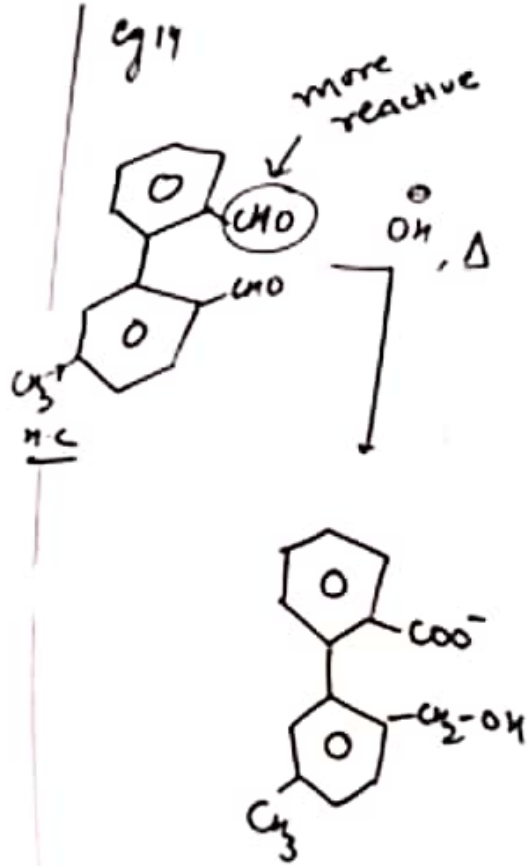
eg 12



eg 13



eg 14



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CARBONYL COMPOUNDS

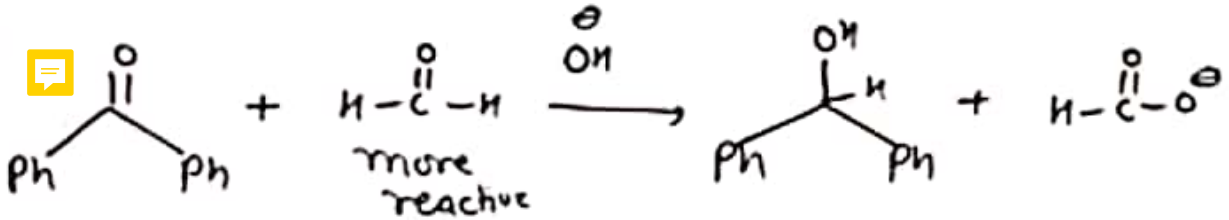
CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

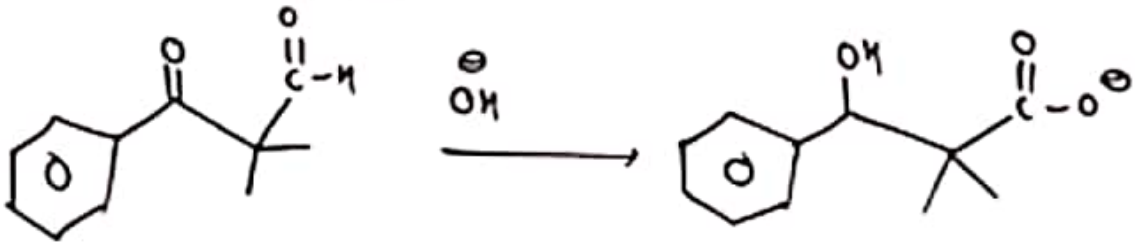
[D1] Cannizzaro Reaction :-

crossed Cannizzaro Rx^n

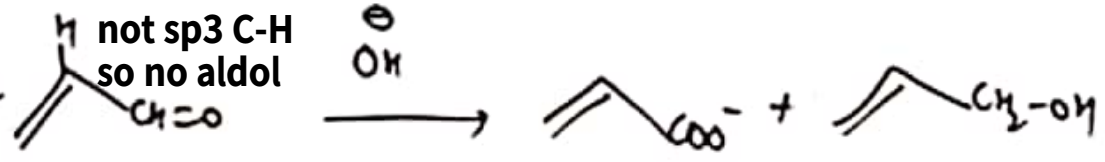
eg 15



eg 16



eg 17



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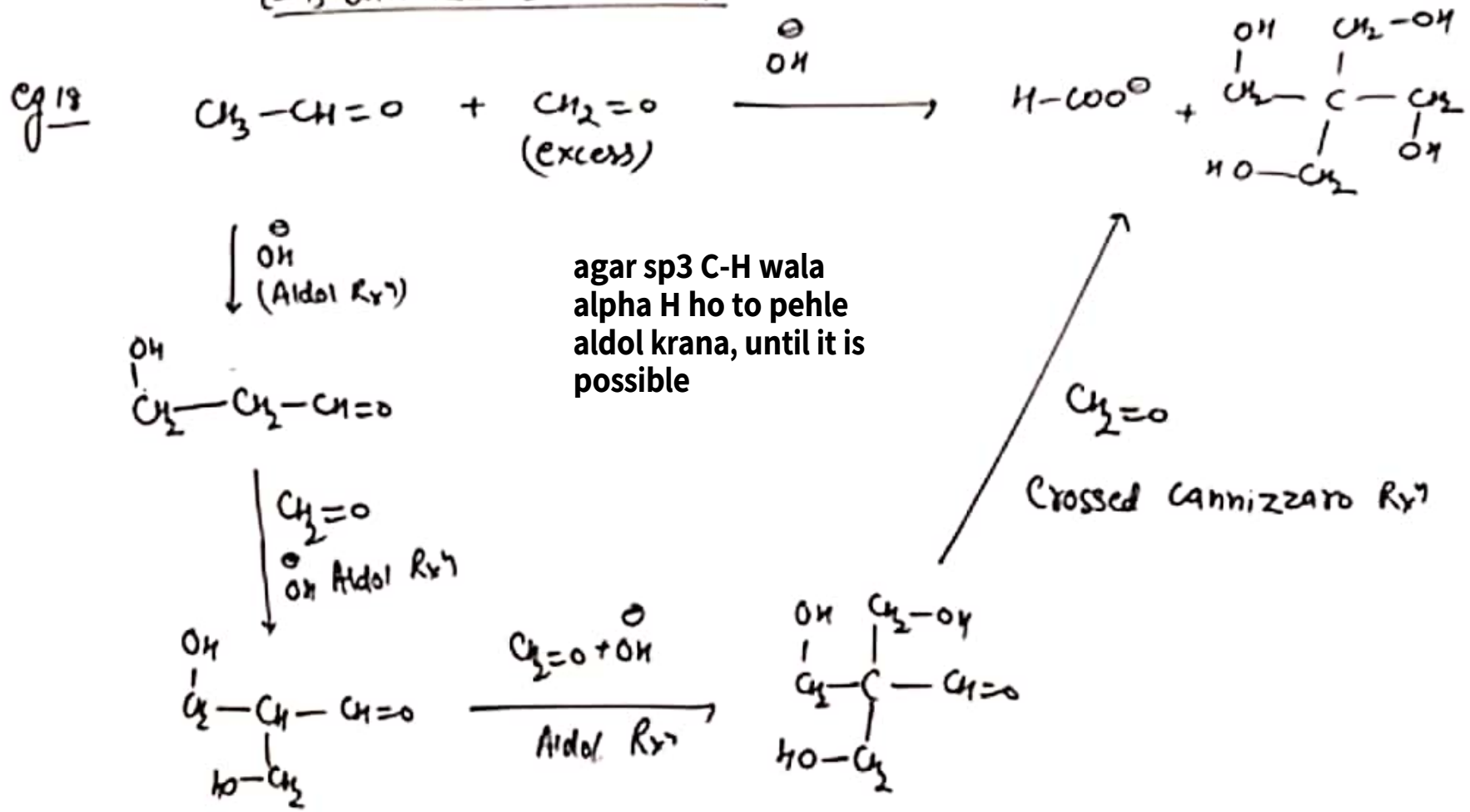
CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

[D1] Cannizzaro Reaction :-

Eg 18 —



agar sp3 C-H wala alpha H ho to pehle aldol krana, until it is possible

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CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

[D1] Cannizzaro Reaction :-

Exception

① $(CH_3)_2CH-CH=O$

$\xrightarrow{\ominus OH}$ ✗ No Aldo Condensation

$\xrightarrow{\ominus OH}$ ✓ Cannizzaro Reaction

② X_3C-CHO (No α-Hydrogen.)

$\xrightarrow{\ominus OH}$ ✗ No Cannizzaro Reaction

$\xrightarrow{\ominus OH}$ ✓ (Haloform Reaction) $CHX_3 + H-COO^-$ Haloform.

experimental fact
due to steric reasons

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CARBONYL COMPOUNDS

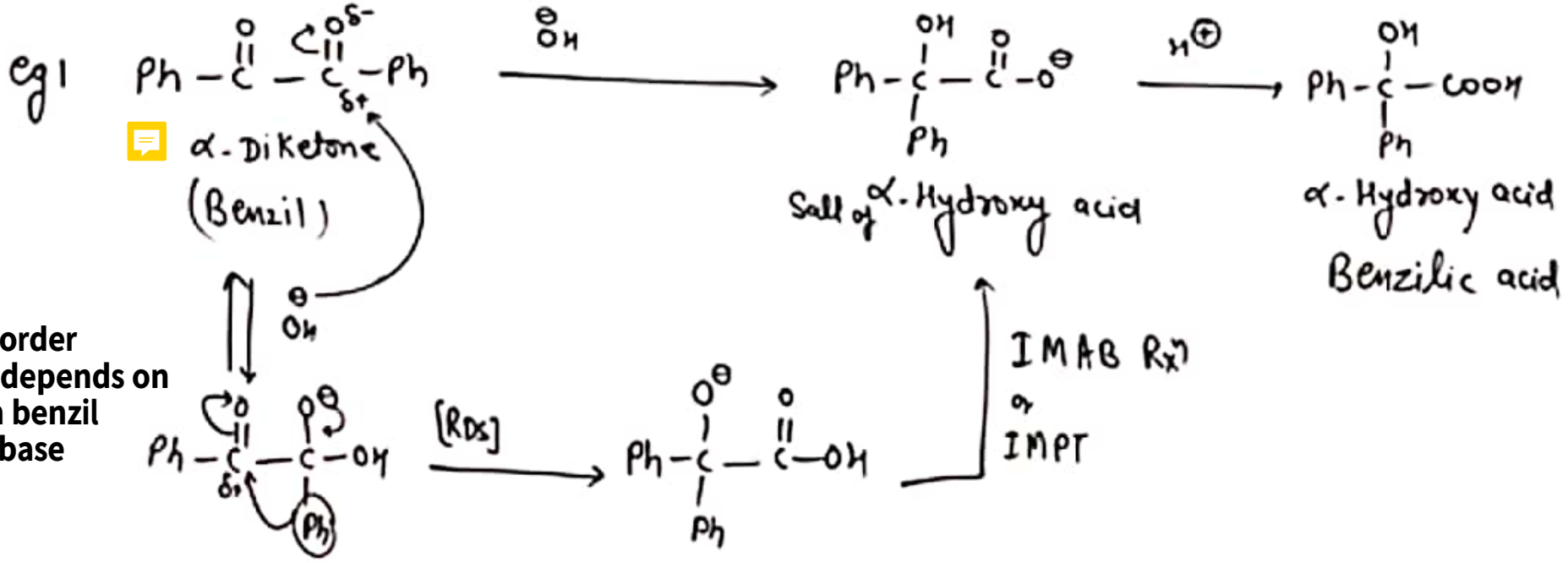
CHEMICAL REACTIONS:-

[D] OTHER REACTIONS:-

[D2] - Benzil - Benzilic acid Rearrangement:-

Adjacent dione in presence of conc base form α-hydroxy acid (salt)

→ Mechanism - Same as intramolecular cannizzaro reaction.



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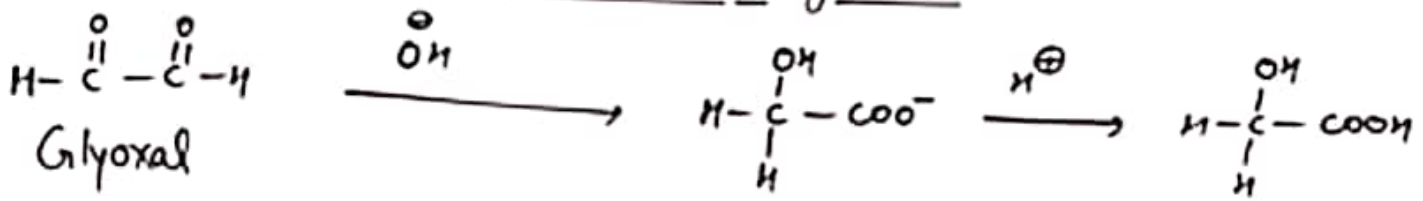
CARBONYL COMPOUNDS

CHEMICAL REACTIONS:-

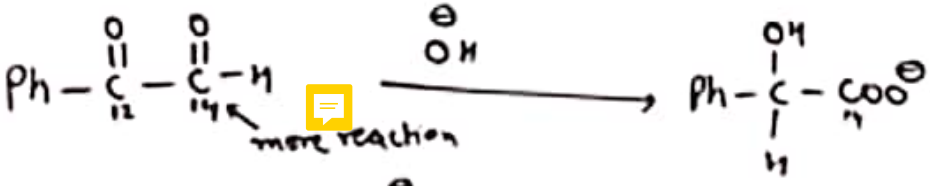
[D] OTHER REACTIONS:-

[D2] - Benzil - Benzilic acid Rearrangement -

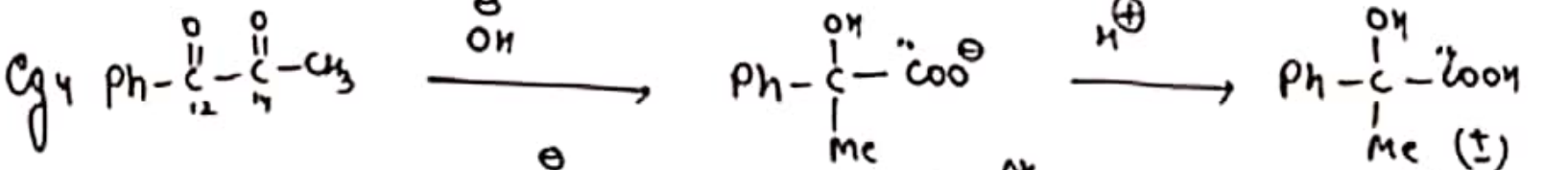
eg 2



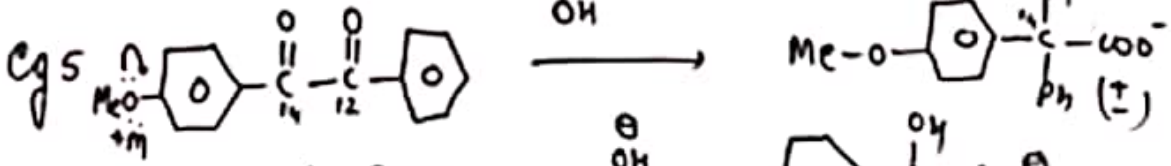
eg 3



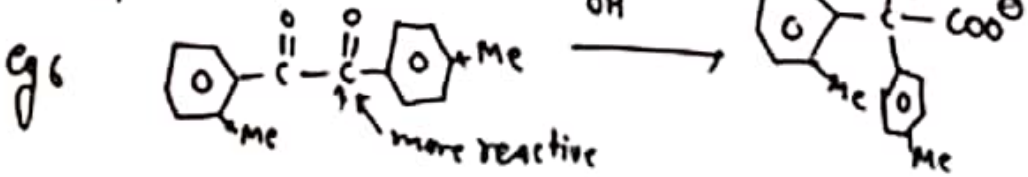
eg 4



eg 5



eg 6



agar C12 and C14 labelling
hi nahi kr rkhi to reactivity ke baare
m sochne ki bhi jarurat nahi
direct bana diya kro
dono side se ek hi product banega

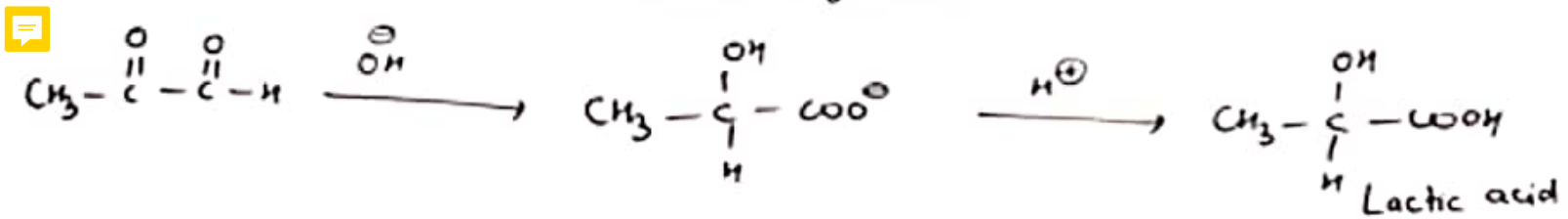
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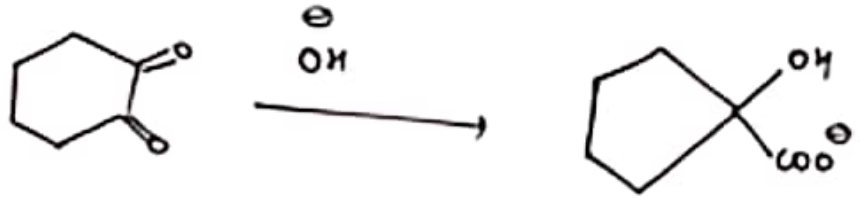
CARBONYL COMPOUNDS

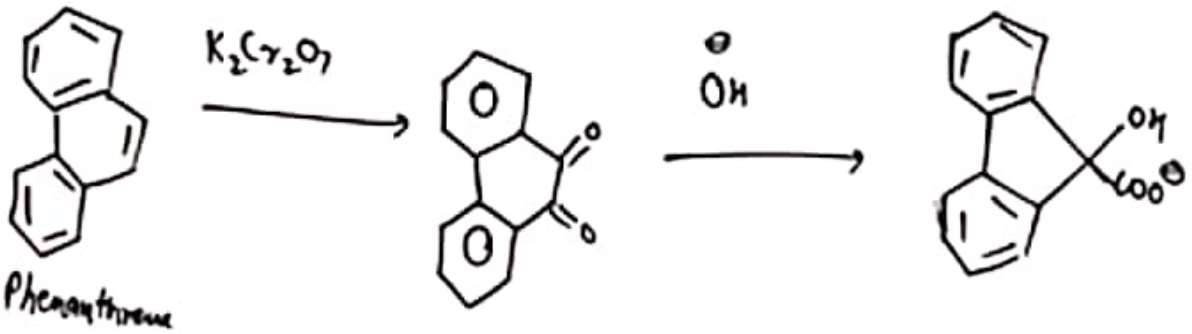
CHEMICAL REACTIONS:-

[D] OTHER REACTIONS:-

[D2] - Benzil - Benzilic acid Rearrangement:-

eg 7 

eg 8 

eg 9 

Semi-bezilic acid rearrangement : OR- in place of OH- I guess, sir named it but said not important

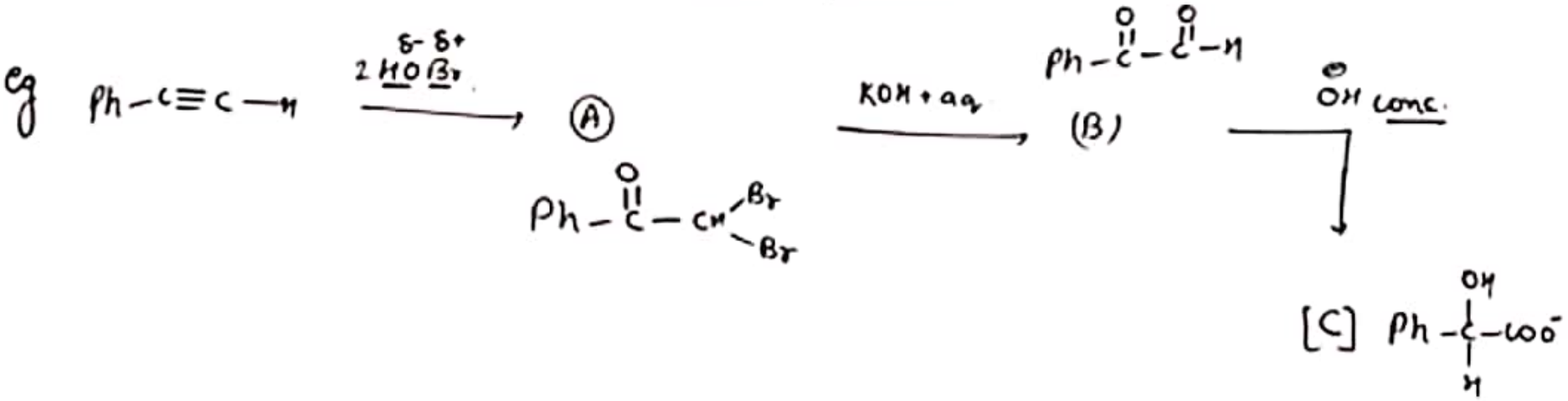
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CARBONYL COMPOUNDS

CHEMICAL REACTIONS:-

[D] OTHER REACTIONS:-

[D2] - Benzil - Benzilic acid Rearrangement:-



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CARBONYL COMPOUNDS

CHEMICAL REACTIONS:-

[D] OTHER REACTIONS:-

[D3]- Tishchenko Reaction:- also called Pseudo-Cannizaro reaction

Reactant → 2 molecules of aldehydes, not given by ketones

Reagent → $Al(OEt)_3$ aluminium alcoxide

Product → Ester

Hint → Cannizzaro + Esterification (short trick)

$$2\,R\text{-}CH{=}O \xrightarrow{Al(OEt)_3} R\text{-}\overset{O}{\overset{\|}{C}}\text{-}O\text{-}CH_2\text{-}R \quad \text{Ester}$$

Scanned with CamScanner

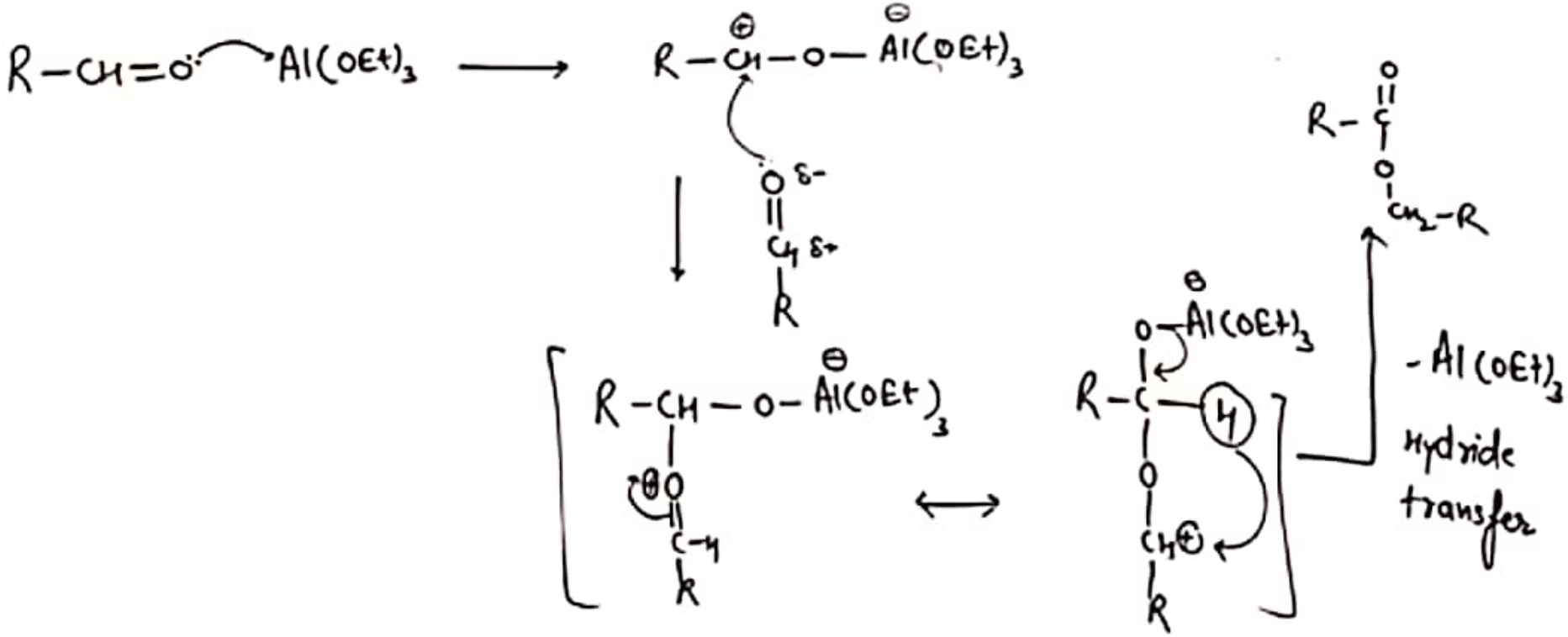
CARBONYL COMPOUNDS

CHEMICAL REACTIONS:-

[D] OTHER REACTIONS:-

[D3]. Tishchenko Reaction:-

Mechanism



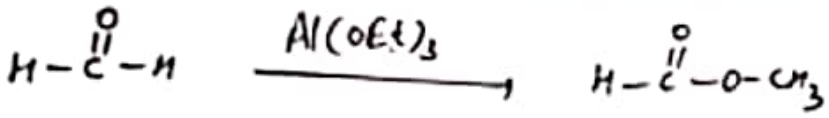
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CARBONYL COMPOUNDS

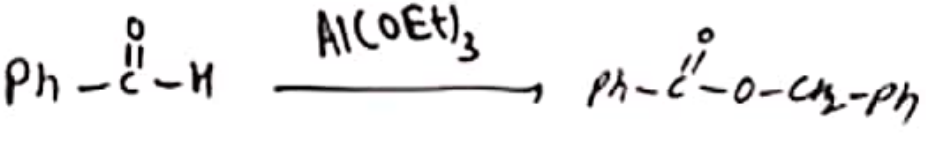
CHEMICAL REACTIONS:-

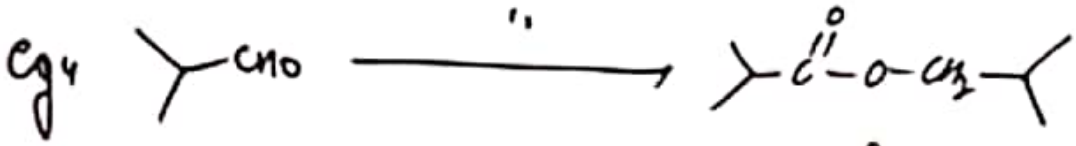
[D] OTHER REACTIONS:-

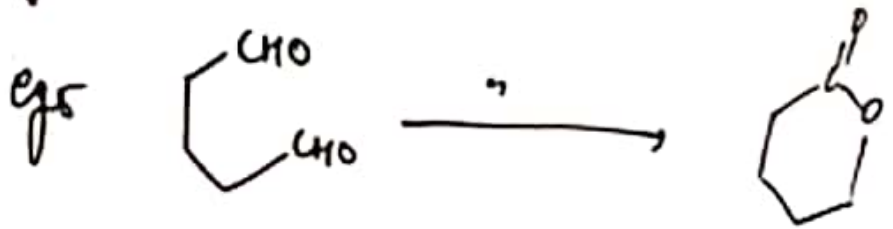
[D3]- Tishchenko Reaction:-

eg 1 

eg 2 

eg 3 

eg 4 

eg 5 

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CARBONYL COMPOUNDS

CHEMICAL REACTIONS:-

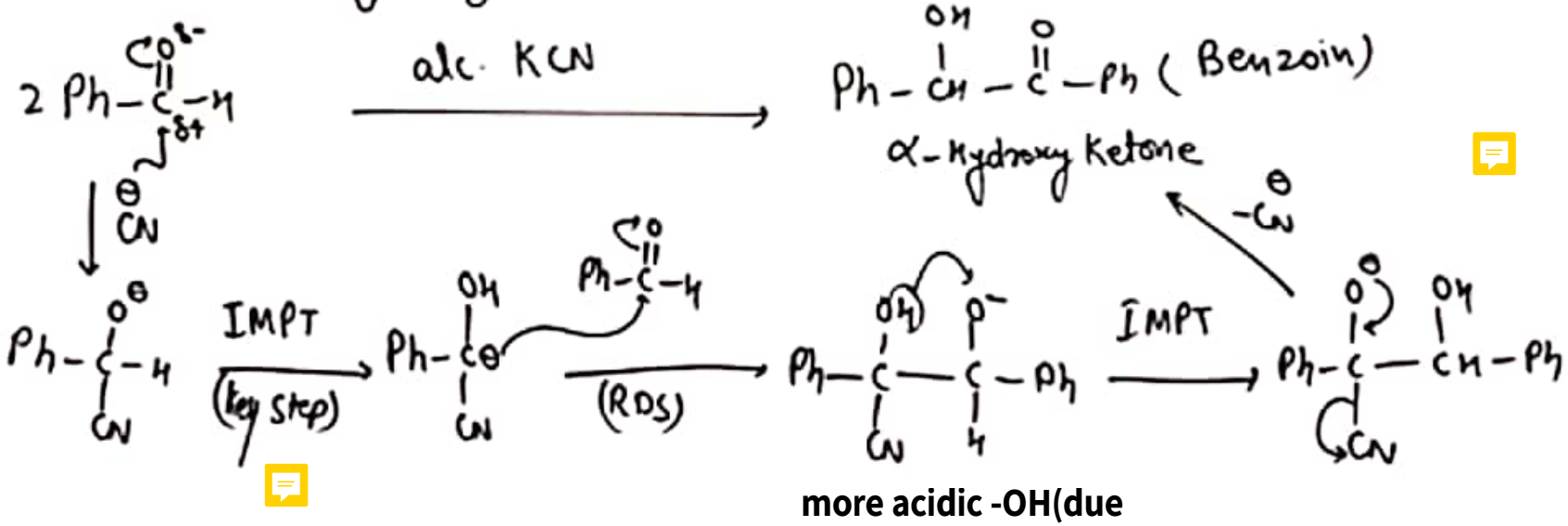
[D] OTHER REACTIONS:-

[D.4] Benzoin condensation:-

Reactant = 2 molecules Aromatic aldehyde **not ketone**

Reagent = alcoholic KCN (aq. KCN forms cyanohydrin; here we take alcoholic)
(KCN ← specific catalyst)

Product = α- Hydroxy Ketone



more acidic -OH(due to -M of -CN)

Scanned with CamScanner

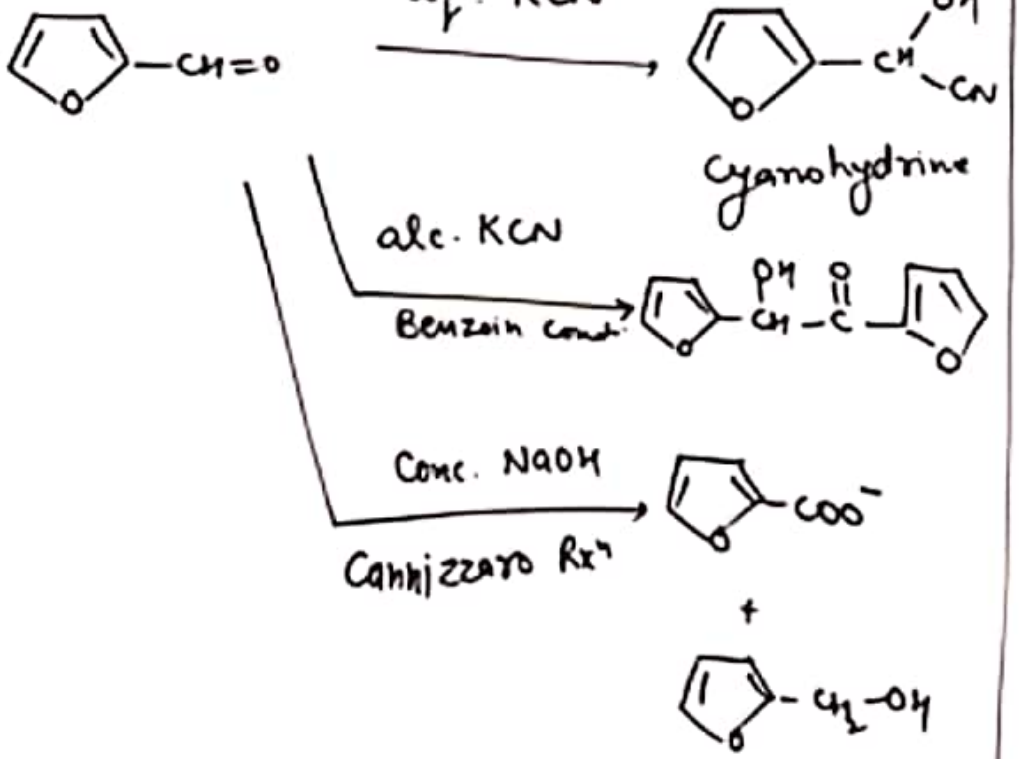
CARBONYL COMPOUNDS

CHEMICAL REACTIONS:-

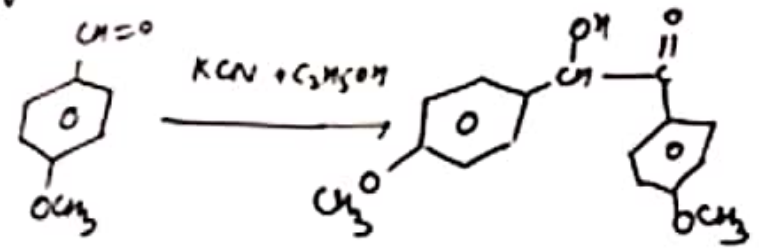
[D] OTHER REACTIONS:-

[D.4] Benzoin condensation:-

eg1



eg2



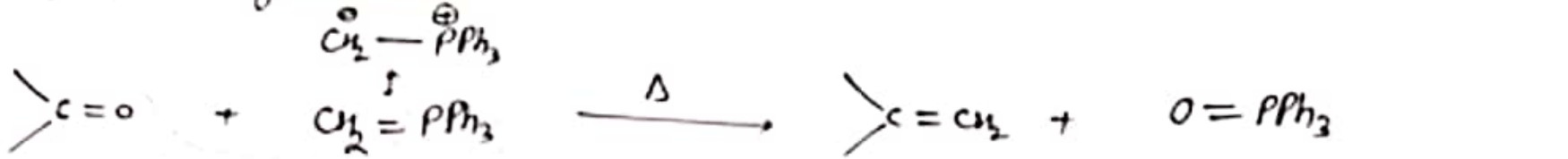
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CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

(D.5) Witting Reactions :-



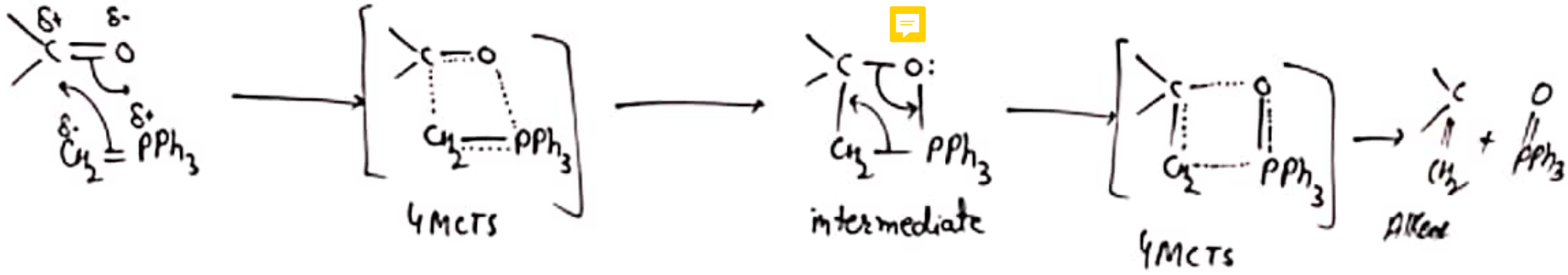
aldehyde or ketone

methylene triphenyl phosphene (ylide)

alkene

Triphenyl phosphoxide

mechanism Pericyclic



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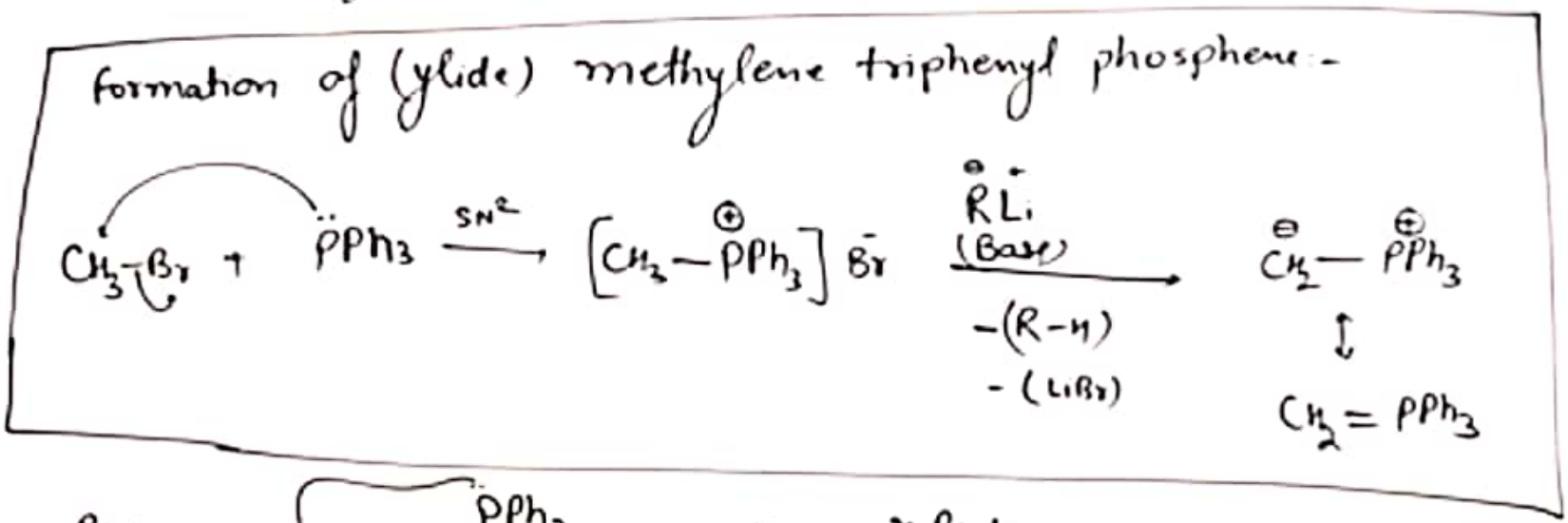
CARBONYL COMPOUNDS

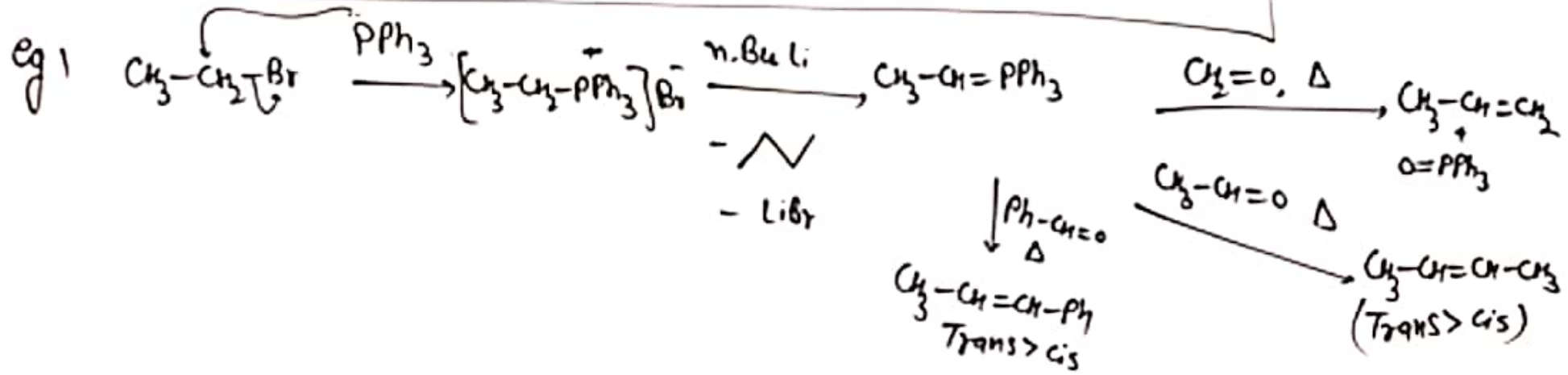
CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

(D.5) Wittig Reactions :-

Formation of (ylide) methylene triphenyl phosphene :-





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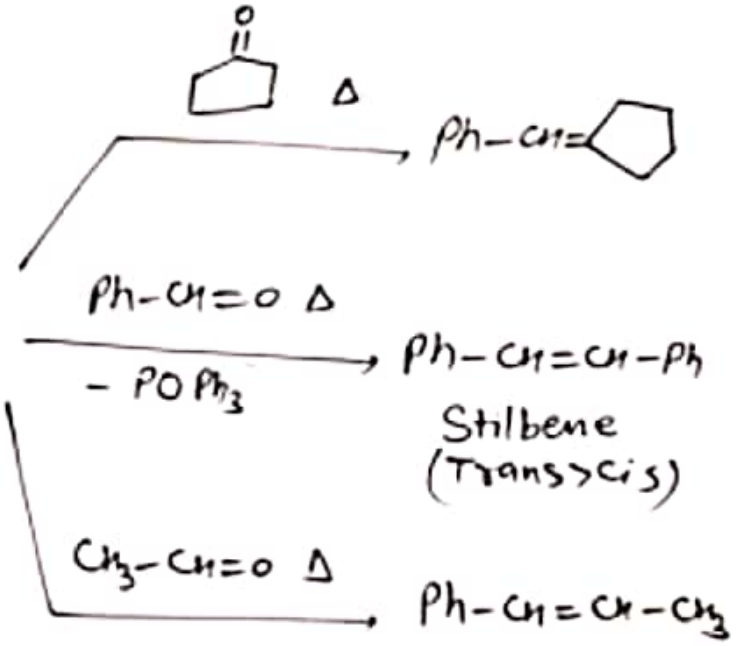
CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

(D.5) Wittig Reactions -

eg 2 $Ph-CH_2-Br \xrightarrow[\text{(ii) n-BuLi}]{\text{(i) } PPh_3} Ph-CH=PPh_3$



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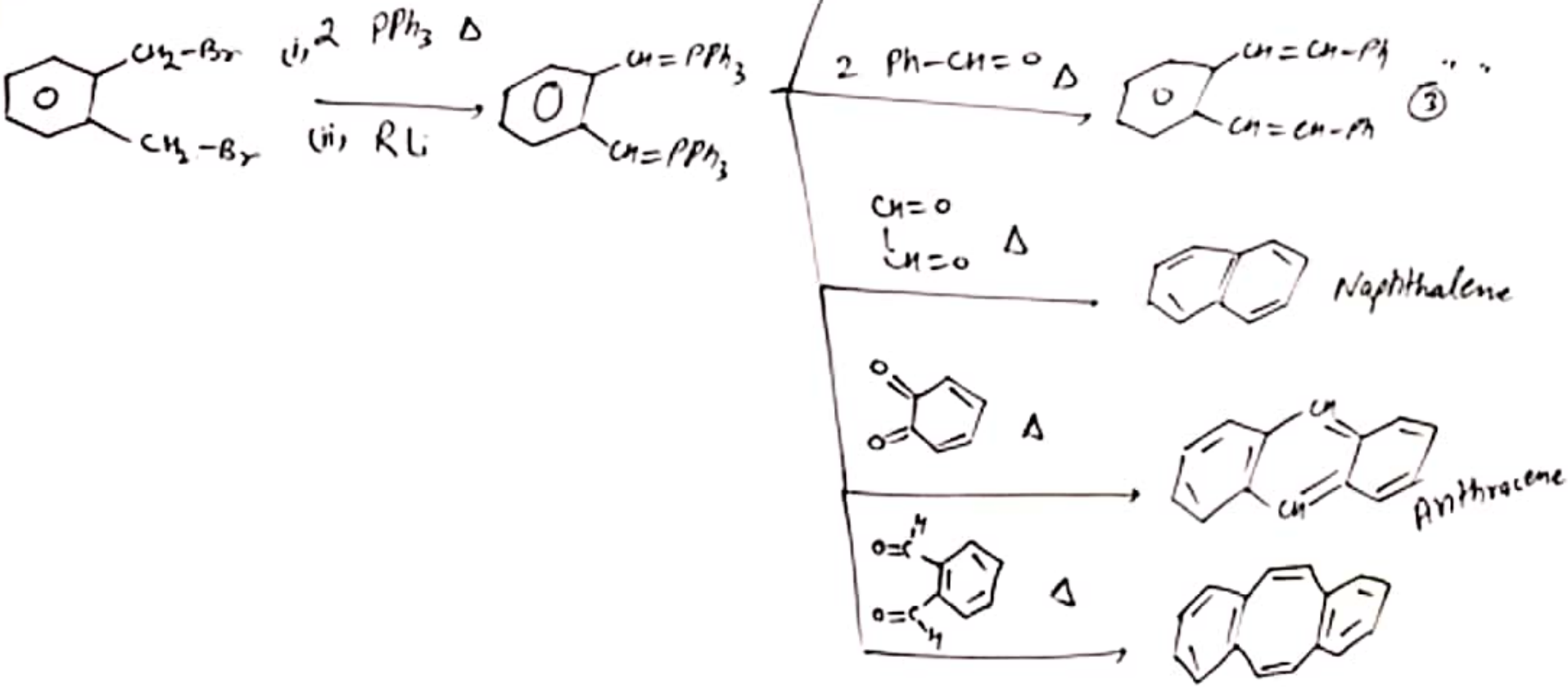
CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

(D.5) Wittig Reactions :-

eg 3



alkene
3 Product
③

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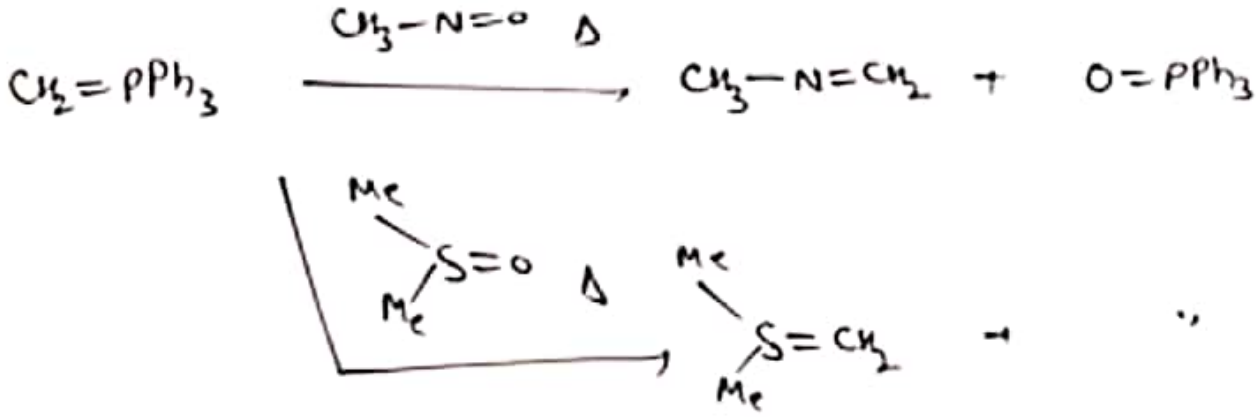
CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

(D.5) Wittig Reactions :-

Eg 4



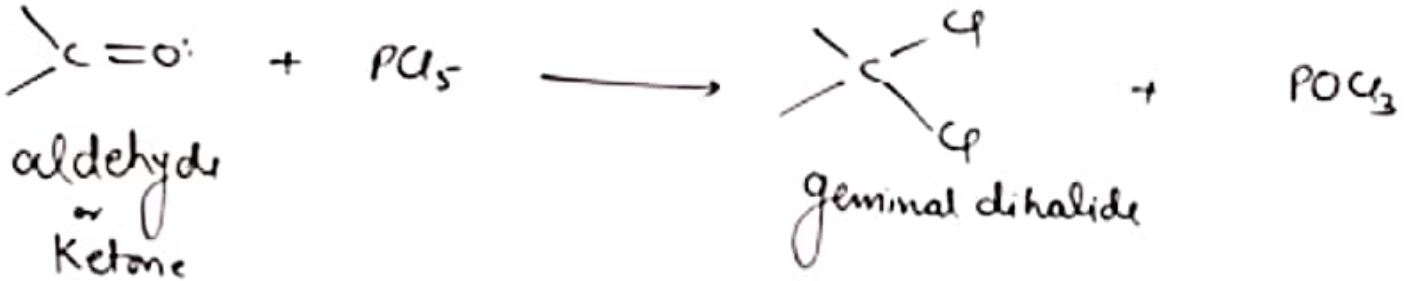
bas P ke saath O nikal do
bache hue ke saath alkene bana do

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CHEMICAL REACTIONS :-

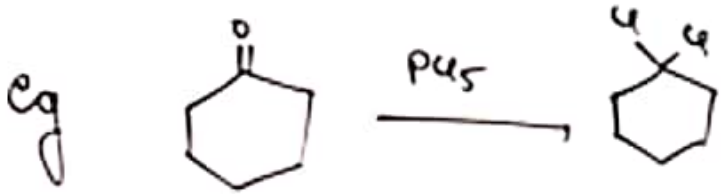
[D] OTHER REACTIONS :-

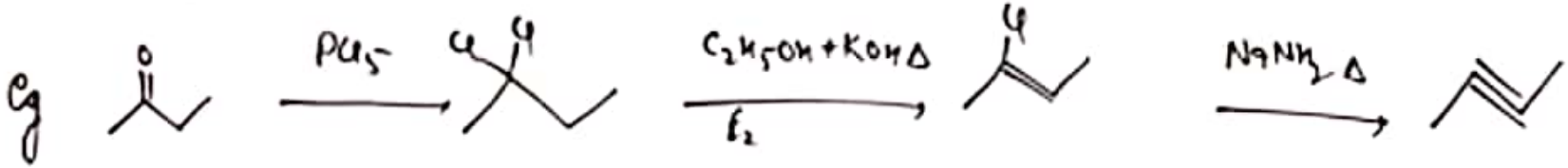
[D6] Reaction with PCl_5



aldehyde or Ketone

geminal dihalide





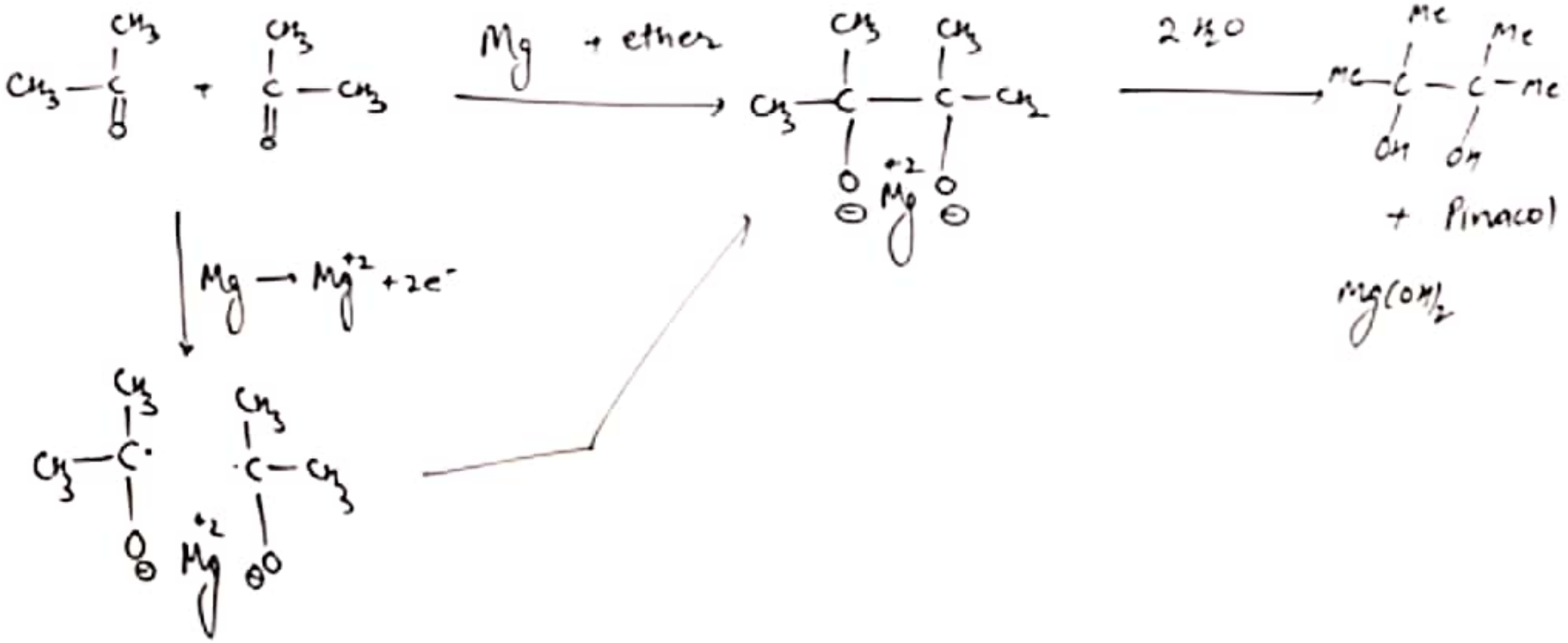
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CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

(D7) Formation of Pinacol (Vic diol)



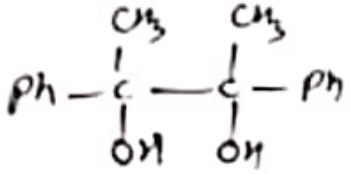
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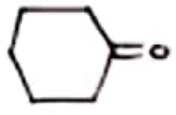
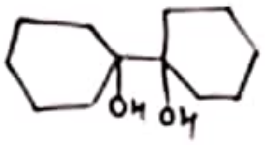
CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

(D7) Formation of Pinacol (Vic diol)

eg1 Acetophenone $\xrightarrow[\text{(ii) } H_2O]{\text{(i) Mg}}$  3 Products

eg2  $\xrightarrow[\text{(ii) } H_2O]{\text{(i) Mg}}$ 

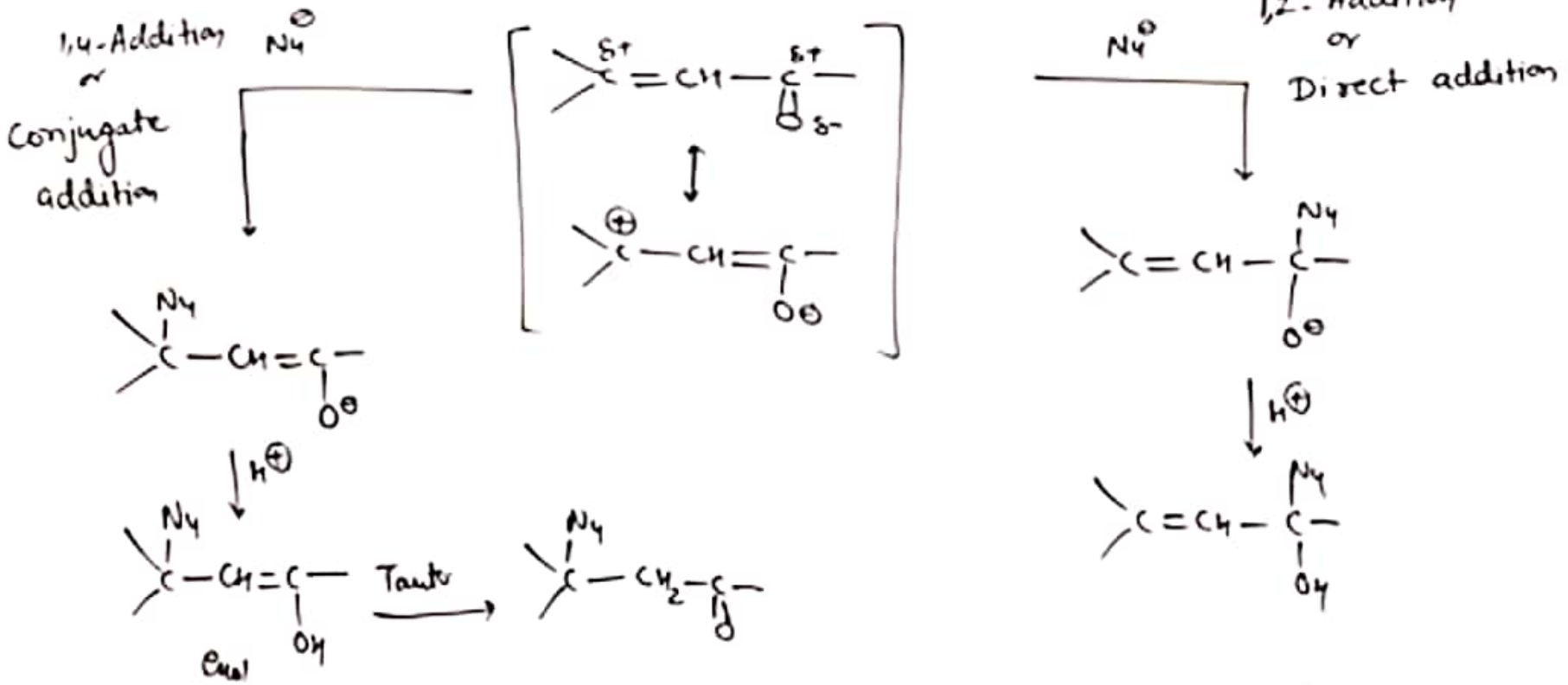
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CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

[D·8] Nucleophilic Addition Reaction on α-β- Unsaturated carbonyl compounds.



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CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

[D·8] Nucleophilic Addition Reaction on α-β· Unsaturated carbonyl compounds

⇒ 1,2 - Addition :- $LiAlH_4$,

smaller and stronger nucleophile

$NaBH_4$

$RMgX$

KCN (for aldehyde) (for ketones; they give 1,4 addition with KCN)

⇒ 1,4 - Addition :- R_2CuLi

R_2Cd

$CH_2=CH-NMe_2$ (Enamine) ↔ $\overset{\ominus}{C}H_2-CH=\overset{\oplus}{N}Me_2$

$\overset{\ominus}{C}H$ — (-ve)

$AlMe_3$ — (-ve) (Michael addition)

→ KCN (for ketone)

→ H-Br

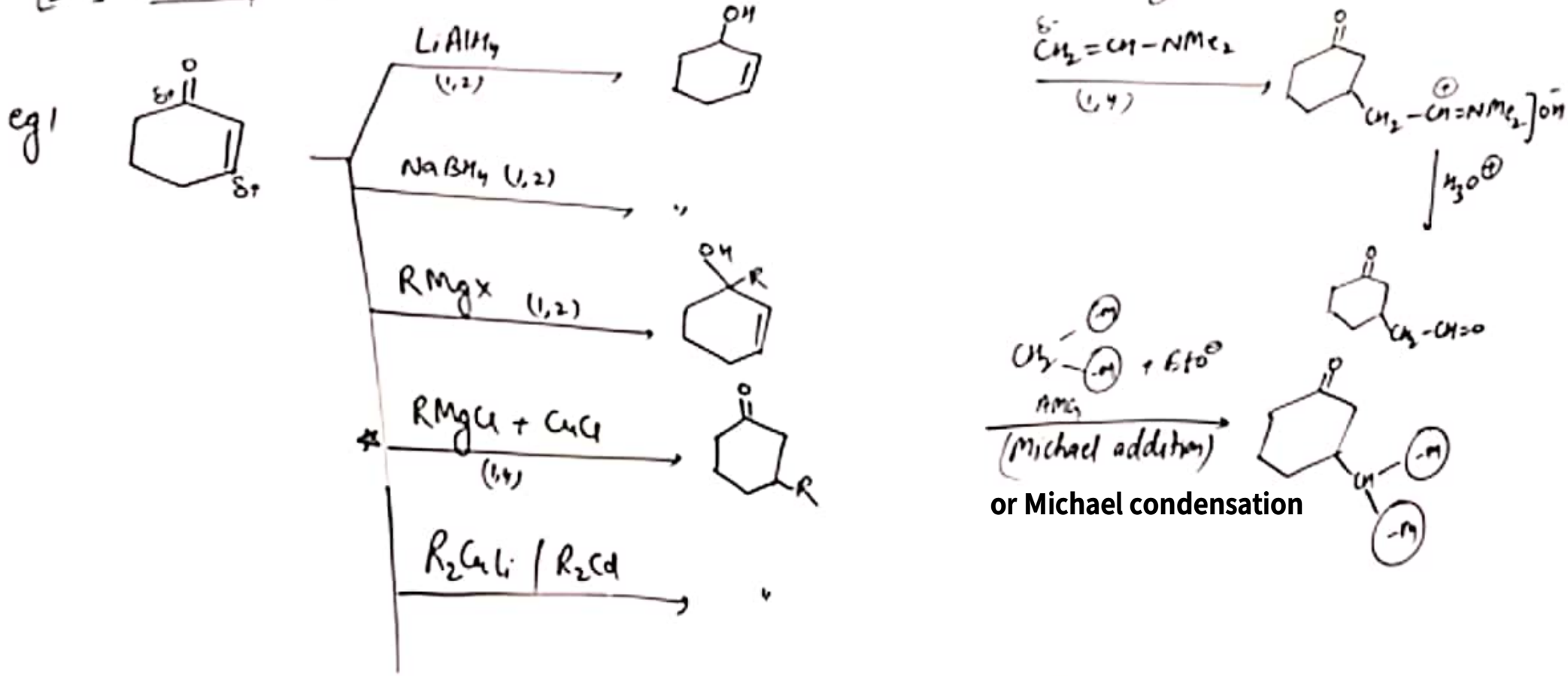
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CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

[D·8] Nucleophilic Addition Reaction on α-β- Unsaturated carbonyl compounds



or Michael condensation

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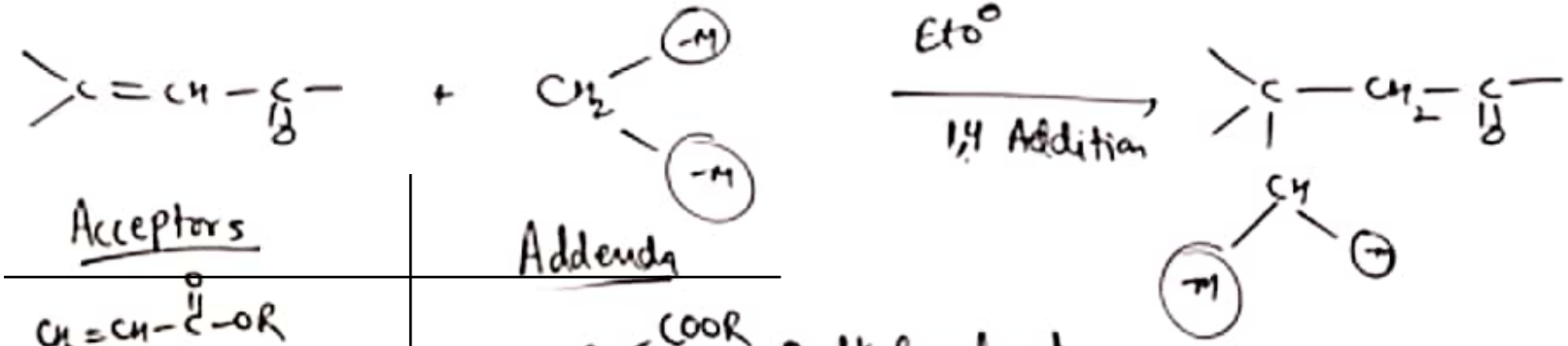
CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

[D 8] Nucleophilic Addition Reaction on α-β- Unsaturated carbonyl compounds.

⇒ Michael condensation:-



Acceptors	Addends
$CH_2=CH-\overset{O}{\overset{\|}{C}}-OR$	$CH_2(COOR)_2$ Dialkyl malonate
$CH_2=CH-CN$	CH_3-COR
$CH_2=CH-\overset{O}{\overset{\|}{C}}-R$	$CH_2(COOR)_2$
$HO-\overset{O}{\overset{\|}{C}}-CH=CH-COOH$	CH_3-NO_2

$CH_2(CN)_2$ etc.

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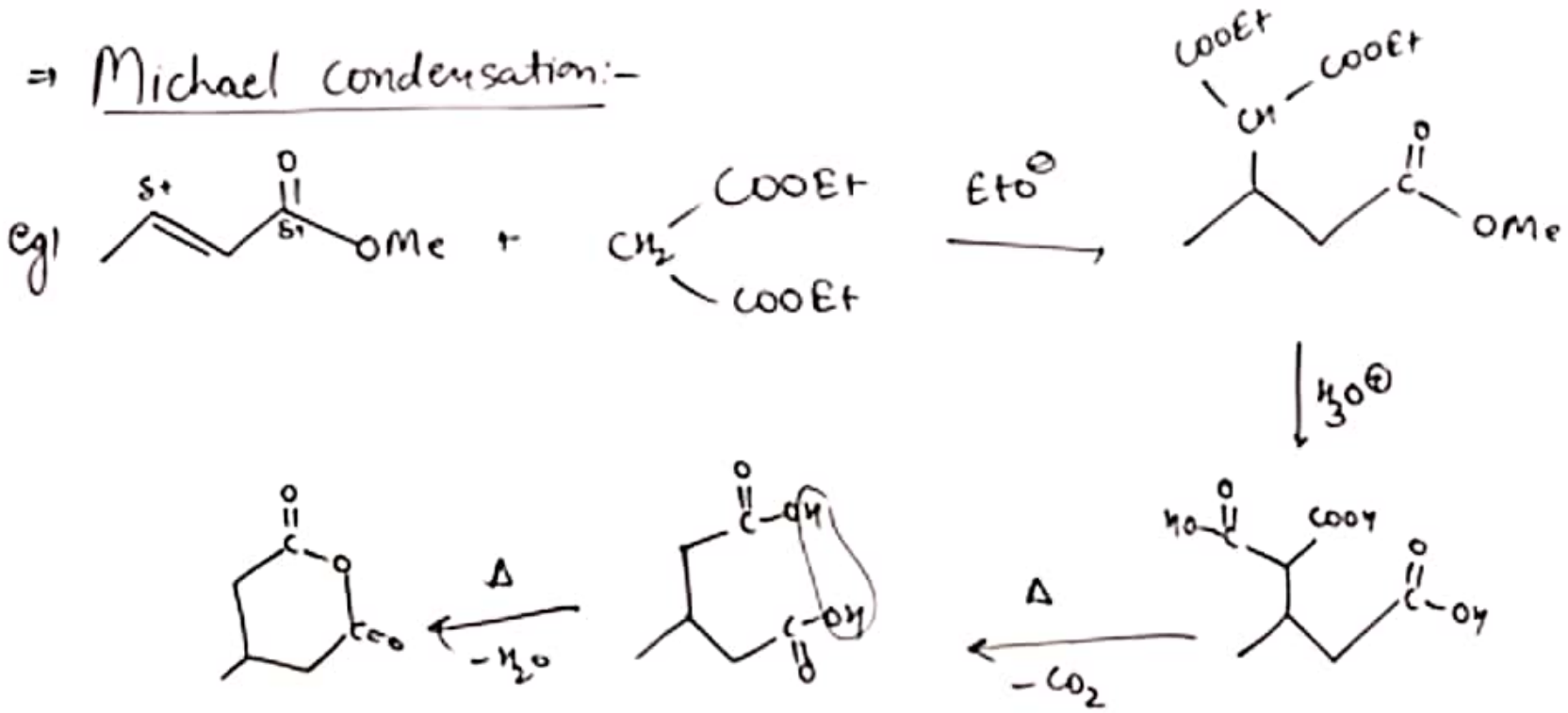
CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

[D] OTHER REACTIONS :-

[D·8] Nucleophilic Addition Reaction on α-β- Unsaturated carbonyl compounds

⇒ Michael condensation:-



Scanned with CamScanner

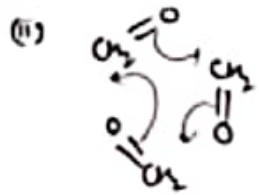
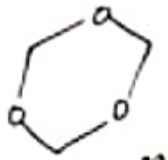
CARBONYL COMPOUNDS

CHEMICAL REACTIONS :-

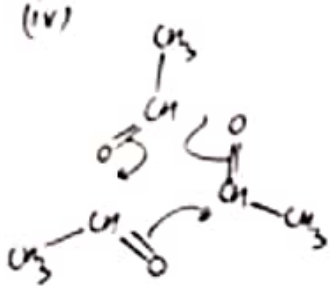
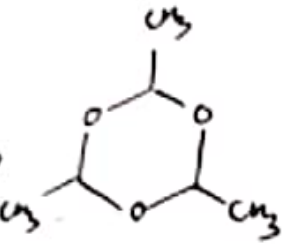
[D] OTHER REACTIONS :-

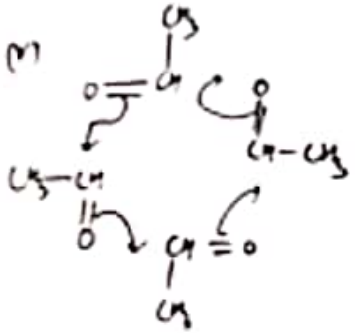
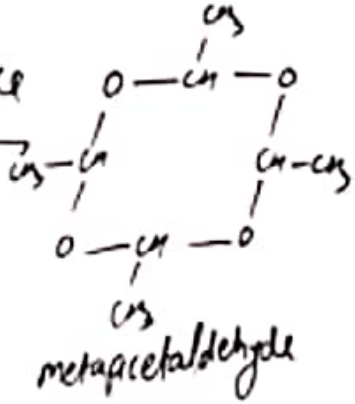
[D9] Polymerisation :-

(i) $CH_2=O$ (40% formalin) $\xrightarrow[\text{evaporate}]{100^\circ C}$ $(CH_2=O)_n H_2O$ or $H(CH_2-O-CH_2-O-CH_2-O)_n$ Paraformaldehyde $\quad n = 6-50$

(ii)  $\xrightarrow{\text{Allowed to Stand}}$  Trioxane or metaformaldehyde

(iii) $6CH_2=O \xrightarrow{Ca(OH)_2\ sol} C_6H_{12}O_6$ Formose Sugar

(iv)  $\xrightarrow[\text{room temp.}]{H_2SO_4}$  Paraacetaldehyde

(v)  $\xrightarrow[0^\circ C]{H_2SO_4 / HCl}$  metaacetaldehyde

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